

Exactly Degenerate Quantum Chaos: Non-Abelian Quantum Geometry as a Probe

Thomas J. Wang^{1,*} and OkongOyangO²

¹*Tsinghua University, Beijing 100084, China*

²*The Pennsylvania State University, University Park, Pennsylvania 16802, USA*

(Dated: August 17, 2026)

Exact degeneracy makes ordinary level statistics spectrally silent: within a protected rank- D manifold the connected energy spectral form factor vanishes identically. It does not, however, fix how the degenerate subspace moves through Hilbert space. We formulate this motion through the non-Abelian quantum geometric tensor of the protected projector and separate protection and mixing: the traceless intrafiber response may vanish while the off-fiber response remains nonzero. We compare two- and three-body fractional-quantum-Hall parents, random and local supersymmetric cohomology, and an X-cube stabilizer control. Finite- D curvature spectra show Jacobi-like local repulsion in several stochastic sectors, whereas longer-range correlations and four-point contractions retain parent- and covariance-dependent memory. For the complete fourth-order test, we subtract the exactly enumerated population mean of each registered finite ensemble, fit the channel whitener on 12 frozen panels, and evaluate 12 held-out panels with all three complex Wick pairings. With a Student- t_{11} reference and a Bonferroni correction over five cases, only Moore-Read $N = 6$ passes the positive directional gate; Moore-Read $N = 4$ and lattice-SUSY ranks $D = 2, 4, 8$ do not. The X-cube control has either zero curvature or deterministic scalar curvature with zero connected variance. These results establish a finite-rank, mechanism-dependent geometric distinction inside spectrally silent manifolds, not an asymptotic or universal Geometric ETH.

I. INTRODUCTION

Quantum chaos is usually inferred from the spectrum. After resolving exact symmetries, nearby levels in a nonintegrable many-body Hamiltonian repel, and longer-range correlations approach the appropriate random-matrix ensemble. The same spectral viewpoint underlies much of the connection between quantum chaos and the eigenstate thermalization hypothesis (ETH) [1, 2]. Wave-function chaos can also appear as universal spectral correlations among reduced-density-matrix eigenvalues, distinct from the energy-level point process [3]. The practical starting point is nevertheless usually a nondegenerate sequence of energies.

An exactly degenerate manifold removes that starting point. Let a rank- D subspace have energy E_0 and remain separated from its complement by a positive gap. Within this subspace every level spacing is zero. Unfolding is undefined, the spacing ratio is trivial, and the contribution of the manifold to the spectral form factor is silent about how its states mix: it depends on D and E_0 , not on the orientation of the subspace in the ambient Hilbert space. This spectral silence does not classify the Hamiltonian as integrable or chaotic. Exact equality of energies can follow from symmetry multiplets, a common kernel of positive constraints, supercharge cohomology, or dependent commuting stabilizers. Each mechanism can coexist with a large and highly structured complement.

The missing information is carried by the motion of the spectral projector. For a smooth family $H(\lambda)$, denote the protected projector by $P(\lambda)$ and its complement by

$Q = 1 - P$. Kato's formulation of adiabatic transport makes P , rather than a basis chosen inside its range, the natural object [4]. The Abelian geometric phase [5], its non-Abelian extension for degenerate multiplets [6], and the quantum metric [7] all arise from derivatives of this projector. The rectangular operator

$$X_a = Q(\partial_a P)P$$

is the off-fiber projector response: it records how a coupling displacement rotates protected states into the positive-energy complement. Products $X_a^\dagger X_b$ form the non-Abelian quantum geometric tensor, and their antisymmetric part gives a curvature acting within the protected fiber. Unlike raw Berry-connection entries, its spectrum and traced contractions are gauge invariant.

Geometry has already proved sensitive to chaos in non-degenerate many-body eigenstates [8]. More directly relevant here, non-Abelian Berry curvature was proposed as a chaos probe for exactly degenerate BPS microstates, where conventional level statistics cannot resolve dynamics inside the protected sector [9]. We ask the corresponding condensed-matter question: what statistical structure can the projector response retain when degeneracy is enforced algebraically, rather than obtained by tuning unresolved levels together? The aim is not to replace spectral chaos by a single geometric number. It is to determine which features of random-matrix behavior survive in the geometry, which remain sensitive to the parent Hamiltonian, and which exact mechanisms force a structured negative control.

Two distinctions are essential. The first separates protection from mixing. For $h_a = \partial_a H$, the traceless intrafiber perturbation

$$S_a = Ph_aP - \frac{\text{Tr}(Ph_aP)}{D}P$$

* WangTheoPhys@outlook.com

tests first-order splitting, whereas X_a tests motion of the subspace. Along an exactness-preserving transport, $S_a = 0$ and the rank and external gap remain fixed, yet X_a can be nonzero. Degeneracy and geometric mixing are therefore independent response channels. This observation also prevents a common false inference: a frustration-free or supersymmetric parent need not have trivial geometry merely because its protected energy is pinned exactly.

The second distinction concerns how a tangent is realized. An exactness-preserving transport is a finite family of constraints or generators for which the common kernel stays exactly degenerate. In the clustered Hall parents studied below, differentiating an invertible transport of the constraint map gives $S_a = 0$ identically. By contrast, the Laughlin fixed-parent base-point probes apply local potentials at a single exact parent. They define the linear response of an isolated rank- D cluster, but a generic finite displacement splits that cluster. We keep these two constructions separate: holonomy and finite-path statements use only the first, while the second supports only base-point response statistics.

Once the spectrum is silent, the statistical question also divides into scales. At the local level we compare curvature eigenvalues with the finite-rank complex Jacobi process associated with products and compressions of random subspaces [10]. Jacobi-like local eigenvalue repulsion is evidence for correlated mixing within one curvature matrix; it does not imply that all response entries are independent Gaussian variables. At the matrix level the two-point covariance and pseudocovariance retain the tangent labels, complement indices, fiber indices, and parent Hamiltonian. At fourth order one must subtract all three complex Wick contractions using that complete covariance. We refer to the surviving dependence of global statistics on this tensor structure as global covariance memory. Local random-matrix-like repulsion and global covariance memory are compatible, and their coexistence is one of the central finite-size observations tested here.

Our comparison is organized by the origin of exact degeneracy. Two-body Laughlin zero modes are realized in the Kapit–Mueller Chern band and in an independent continuum lowest-Landau-level construction. Three-body Moore–Read zero modes change the clustering constraint and greatly enlarge the protected rank. A random cubic supercharge and a local lattice $\mathcal{N} = 2$ supercharge realize two inequivalent cohomological kernels. The periodic X-cube stabilizer code supplies an exactly soluble structured control. Topological stability under gapped local deformations motivates the use of projector transport in the Hall setting [11], but topology alone does not determine the local curvature statistics. The models therefore vary the protection algebra, the locality of the tangent, and the accessible response channels rather than serving as interchangeable realizations of one ensemble.

The corrected fourth-order comparison is narrower than the model list. After exact population centering for each registered finite ensemble, channel whitening on a frozen

12-panel training split, and complete three-pairing Wick subtraction on 12 held-out panels, only the registered finite-rank Moore–Read $N = 6$ case passes the positive directional gate under a Student- t_{11} reference with a Bonferroni correction over five primary cases. Moore–Read $N = 4$ and the lattice-SUSY ranks $D = 2, 4, 8$ do not pass. No corresponding centered full-cumulant statement is assigned to the Laughlin or random-SYK production data. This pattern establishes neither Gaussian closure in the nonpassing cases nor a common ensemble across mechanisms; it fixes the claim boundary of the global comparison.

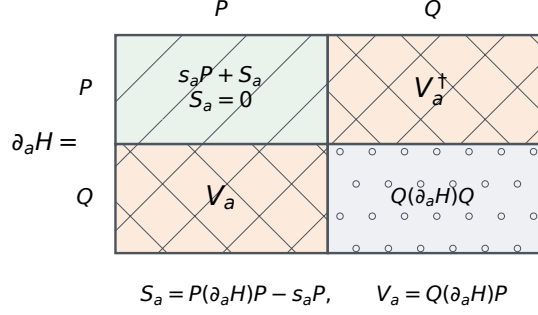
The resulting claim is deliberately finite. We find finite-rank, cross-mechanism evidence that exact spectral degeneracy can coexist with an informative and, in several stochastic sectors, locally Jacobi-like quantum geometry. The response nevertheless remembers its constraint algebra and operator class through its covariance and branch structure. The X-cube control makes the distinction sharp: coefficient deformations have zero projector curvature, while a chosen unitary transport has deterministic curvature whose connected variance vanishes. Thus exact degeneracy alone does not produce geometric chaos.

These calculations do not constitute an asymptotic Geometric ETH theorem. They neither establish a common scaling limit nor identify a single model/operator class that can be held fixed across all mechanisms. The complete-covariance inference is accepted only for the registered Moore–Read $N = 6$ cell whose statistical-unit and familywise gates pass; nonpassing and untested cells are not promoted to conclusions. Section II develops the protected response formalism, Sec. III fixes the hierarchy of statistics and inference units, and Sec. IV defines the exact mechanisms and tangent classes. The remaining sections present the mechanism-resolved calculations, the topological comparison, and the limits of the geometric interpretation.

II. PROTECTED FIBERS AND OFF-FIBER RESPONSE

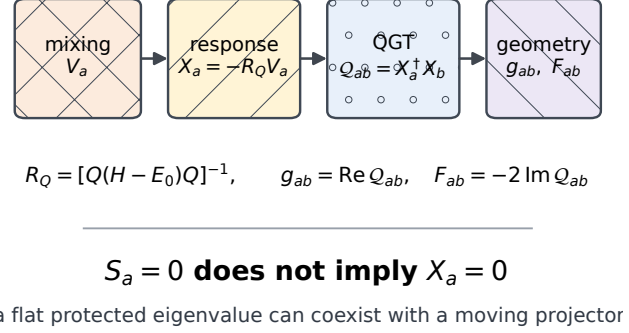
Let $H(\boldsymbol{\lambda})$ be a differentiable family of self-adjoint Hamiltonians. Throughout this paper, $E_0(\boldsymbol{\lambda})$ denotes an isolated eigenvalue of constant multiplicity D on the parameter region under consideration, and $P(\boldsymbol{\lambda})$ is its constant rank D spectral projector. We write $Q = 1 - P$ and assume a nonzero spectral gap $\Delta = \text{dist}[E_0, \text{spec}(QH Q)]$. These assumptions are stronger than observing a numerically narrow band: they ensure that the projector is differentiable and that the inverse of $H - E_0$ on QH is bounded. No response formula below is continued through a point at which the rank changes or the spectral gap closes.

(a) Protection constrains the intrafiber block



exact degeneracy: $S_a = 0$; off-fiber mixing V_a is unrestricted

(b) Off-fiber mixing determines geometry



(c) Audited response classes under exact degeneracy

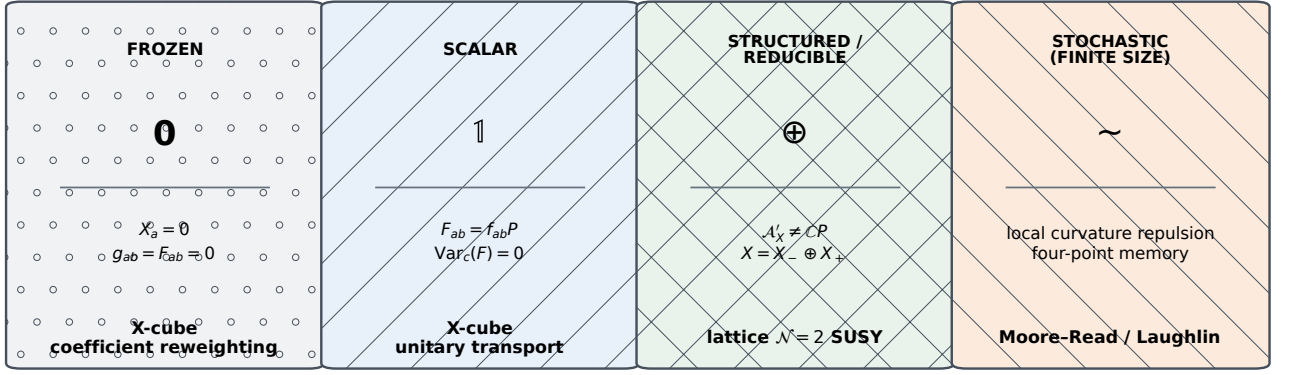


FIG. 1. Protection and mixing channels under exact degeneracy. (a) The tangent $\partial_a H$ is decomposed with respect to the protected projector P and its complement Q . Exact first-order protection constrains the traceless intrafiber block $S_a = P(\partial_a H)P - s_a P$, where $s_a = D^{-1} \text{Tr}[P(\partial_a H)P]$, but leaves the off-fiber mixing $V_a = Q(\partial_a H)P$ unrestricted. (b) The reduced resolvent converts this mixing into the Kato response $X_a = -R_Q V_a$, from which $Q_{ab} = X_a^\dagger X_b$, g_{ab} , and F_{ab} follow; hence $S_a = 0$ does not imply $X_a = 0$. (c) The audited responses divide into frozen, scalar, structured/reducible, and stochastic finite-size classes. X-cube coefficient reweighting and unitary transport realize the first two classes, the local lattice $\mathcal{N} = 2$ supercharge resolves into exact/coexact branches, and the clustered Hall parents provide the stochastic examples. The classification concerns projector response and does not assign level statistics or thermalization to the protected manifold.

A. The splitting and mixing blocks

For a coordinate λ^a , set $h_a = \partial_a H$. The two blocks of h_a that are relevant to the protected fiber are

$$S_a = Ph_a P - \frac{\text{Tr}(Ph_a P)}{D} P, \quad V_a = Qh_a P. \quad (1)$$

The Hermitian operator S_a is the traceless intrafiber splitting channel, whereas V_a maps the protected fiber into its orthogonal complement. If the eigenvalue remains exactly D -fold degenerate along the path, differentiating $HP = E_0 P$ and projecting with P on both sides gives

$Ph_a P = (\partial_a E_0)P$, hence $S_a = 0$. This is a local necessary condition for an exactness-preserving path. It is not, by itself, a claim that an arbitrary linear perturbation with $S_a = 0$ preserves the degeneracy at finite amplitude; the finite path must continue to satisfy the rank and gap conditions.

Crucially, $S_a = 0$ imposes no corresponding condition on V_a . Exact degeneracy fixes the restriction of the Hamiltonian to $P\mathcal{H}$, but it does not fix how that subspace is embedded in the full Hilbert space. This distinction also works in the opposite direction. An intervention supported entirely in $P\mathcal{H}$ can make S_a nonzero, and can

even produce random-matrix level statistics inside the multiplet, while keeping the projector fixed. Such an intervention changes intrafiber energies but produces no quantum geometry.

B. Reduced response of an isolated fiber

Under the assumptions above, the reduced resolvent

$$R_Q = Q[Q(H - E_0)Q]^{-1}Q$$

exists and satisfies $\|R_Q\| \leq \Delta^{-1}$. Projecting the derivative of $HP = E_0P$ from $P\mathcal{H}$ to $Q\mathcal{H}$ gives the off-fiber response

$$X_a = Q(\partial_a P)P = -R_Q V_a. \quad (2)$$

Thus a large response can arise either from a large coupling matrix element V_a or from a small excitation denominator. The distinction matters when systems with different gaps are compared: an unnormalized response does not, by itself, measure more effective mixing channels. Equation (2) also makes the domain of the calculation explicit. It is an adiabatic response of an isolated spectral subspace, not a prescription for following a collection of levels after that subspace touches the rest of the spectrum.

The matrix-valued quantum geometric tensor is the operator on $P\mathcal{H}$

$$\mathcal{Q}_{ab} = P(\partial_a P)Q(\partial_b P)P = X_a^\dagger X_b, \quad g_{ab} = \frac{\mathcal{Q}_{ab} + \mathcal{Q}_{ba}}{2}. \quad (3)$$

The diagonal $g_{aa} = X_a^\dagger X_a$ is positive semidefinite, and its rank counts the fiber directions resolved by tangent a , subject to the gap weighting in R_Q . Unlike the scalar quantum metric of a nondegenerate state, the off-diagonal g_{ab} is generally a matrix inside the protected multiplet; no analogous rank interpretation is assigned to it.

Choose a local orthonormal frame $\Psi = (|u_1\rangle, \dots, |u_D\rangle)$ for $P\mathcal{H}$. With the canonical convention $A_a = i\Psi^\dagger \partial_a \Psi$, the non-Abelian Berry curvature is

$$\begin{aligned} F_{ab} &= \partial_a A_b - \partial_b A_a - i[A_a, A_b] \\ &= i(\mathcal{Q}_{ab} - \mathcal{Q}_{ba}) = iP[\partial_a P, \partial_b P]P. \end{aligned} \quad (4)$$

For $D = 1$, this convention reduces to $F_{ab} = -2\text{Im } \mathcal{Q}_{ab}$. Equations (3) and (4) are therefore the symmetric and antisymmetric parts of the same off-fiber response.

C. Gauge covariance and observable contractions

A change of frame $\Psi \mapsto \Psi U(\lambda)$ leaves the projector unchanged but sends

$$\begin{aligned} A_a &\mapsto U^\dagger A_a U + iU^\dagger \partial_a U, \\ F_{ab} &\mapsto U^\dagger F_{ab} U, \quad \mathcal{Q}_{ab} \mapsto U^\dagger \mathcal{Q}_{ab} U. \end{aligned} \quad (5)$$

The connection transforms inhomogeneously, while the curvature and QGT are gauge covariant. Raw entries of A_a are consequently not primary observables in our statistical tests. The eigenvalues of a fixed Hermitian curvature component, traces such as $\text{Tr } F_{ab}^k$, and closed words such as $\text{Tr}(\mathcal{Q}_{ab}\mathcal{Q}_{cd})$ are gauge invariant. When rectangular response matrices are used computationally, their fiber-frame representation transforms as $X_a \mapsto X_a U$; every reported contraction closes the fiber indices and has the corresponding invariant or covariant transformation law.

This establishes the separation used throughout the paper. The condition $S_a = 0$ protects the eigenvalue to first order, whereas V_a and the reduced resolvent determine the motion of the protected fiber. A fixed projector has $X_a = g_{ab} = F_{ab} = 0$. A moving projector can instead have scalar, block-reducible, or fully resolved curvature. Which of these cases displays random-matrix correlations is a statistical question, not a consequence of the degeneracy alone.

III. STATISTICAL TESTS OF GEOMETRIC CHAOS

No single statistic is sufficient to identify chaos in a degenerate fiber. We therefore use a hierarchy of tests. Exact energy degeneracy supplies the null result at the spectral level. Curvature eigenvalues test local and mesoscopic correlations of one geometric component. Response trace words then test whether the joint distribution of several physical tangent channels is fixed by its complete two-point structure. The hierarchy prevents a local level-repulsion result from being promoted into a statement about Gaussian closure or a thermodynamic limit.

A. Spectral silence and curvature spectra

For a protected multiplet, shift the common energy E_0 separately in each sample and define

$$Z_E(t) = \text{Tr}_P e^{-it(H-E_0)}.$$

Exact degeneracy gives $Z_E(t) = D$. With the normalization used here,

$$\begin{aligned} K_{E,\text{raw}}(t) &= D^{-1} \mathbb{E}|Z_E(t)|^2 = D, \\ K_{E,c}(t) &= D^{-1} (\mathbb{E}|Z_E(t)|^2 - |\mathbb{E}Z_E(t)|^2) = 0. \end{aligned} \quad (6)$$

The equalities are algebraic and contain no dynamical inference. In particular, an energy ramp can be manufactured only by splitting the protected multiplet or by mixing it with states outside the isolated fiber.

The curvature retains a nontrivial spectrum. For a registered tangent pair (a, b) , we remove its scalar part and fix its scale by

$$\tilde{F}_{ab} = F_{ab} - D^{-1}(\text{Tr } F_{ab})P, \quad \sigma_{ab}^2 = D^{-1} \text{Tr}(\tilde{F}_{ab}^2).$$

Samples with $\sigma_{ab} = 0$ are recorded as scalar controls rather than silently normalized. Otherwise $\hat{F}_{ab} = \hat{F}_{ab}/\sigma_{ab}$ has a dimensionless spectrum. We compare its adjacent-gap ratios, connected form factor, and number variance with finite-dimensional reference ensembles. For example, if $Z_F(\tau) = \text{Tr } e^{-i\tau \hat{F}_{ab}}$, then

$$K_{F,c}(\tau) = D^{-1} \left(\mathbb{E} |Z_F(\tau)|^2 - |\mathbb{E} Z_F(\tau)|^2 \right).$$

The expectation is over the declared ensemble, not over eigenvalues of one matrix. Adjacent gaps within a common spectrum are correlated observations; uncertainty intervals are therefore formed by resampling complete matrices or complete registered panels.

Local repulsion and a finite- D form-factor ramp test correlations within one curvature component. They do not test the joint distribution of different tangent directions. Nor do they determine whether deviations at longer scales follow from a nontrivial covariance, a low-rank response algebra, or a connected higher moment. Those questions require matrix-element contractions.

B. Channel whitening and the four-point tensor

Let $a = 1, \dots, M$ label a physical tangent family fixed before examining the outcome. At a common base projector, set $Y_a = X_a - \mathbb{E} X_a$ and define its channel Gram matrix

$$G_{ab} = D^{-1} \mathbb{E} \text{Tr}(Y_a^\dagger Y_b).$$

We whiten on the registered support of G , using its Moore–Penrose inverse when an exact null channel has been identified independently of the fourth moment. The resulting responses are denoted $\hat{X}_a$. Channel whitening removes an overall choice of tangent coordinates, but it does not assume that covariance factorizes between tangent, complement, and fiber indices.

For the Moore–Read and lattice-supercharge panels, the mean is not fitted on the inference sample. It is evaluated exactly for the finite registered tangent population: the uniform torus average over all guiding-center anchors for Moore–Read, and the enumerated ordered local-coordinate marginals for the lattice supercharge. Panels 0–11 determine only the 2×2 channel whitener; the frozen map is applied to panels 12–23, which form the primary inference set. The reversed split is recorded as a descriptive replication and is not pooled with the primary estimate.

The gauge-invariant four-channel tensor is

$$T_{abcd} = \frac{1}{D} \mathbb{E} \text{Tr} \left(\hat{X}_a^\dagger \hat{X}_b \hat{X}_c^\dagger \hat{X}_d \right). \quad (7)$$

A Gaussian comparison must reproduce the complete second moments of the vectorized response. If z stacks all complement, fiber, and channel indices, these moments are the covariance $C = \mathbb{E}(zz^\dagger)$ and the pseudocovariance

$\Pi = \mathbb{E}(zz^T)$. The latter need not vanish for a complex response ensemble. We write their channel-indexed blocks as C_{ab} and Π_{ab} , with all ambient indices retained inside each block. Isserlis’ theorem therefore contains three Wick pairings. Suppressing the contracted ambient indices, their tensor structure is

$$W_{abcd} = [C_{ab}C_{cd}] + [\Pi_{ac}^* \Pi_{bd}] + [C_{ad}C_{cb}].$$

Keeping only the first and third terms imposes a proper-complex null that the physical tangent ensemble need not satisfy. Likewise, replacing each bracket by a product of scalar channel covariances imposes an additional separability assumption. Neither restriction is part of the complete Gaussian comparison.

We define the connected fourth-order response by

$$\mathcal{K}_{abcd} = T_{abcd} - W_{abcd}, \quad R_4^{\text{full}} = \frac{\|\mathcal{K}\|_F}{\|W\|_F}. \quad (8)$$

The ratio is reported only when $\|W\|_F$ passes the registered nonzero-scale gate. A scalar projection of $\mathcal{K}$ can be more powerful than its norm, but its direction and sign must be fixed without reference to the opened data. A statistically resolved nonzero population contraction of $\mathcal{K}$ rejects the matched Gaussian null for the stated ensemble and tangent class; a nonzero finite-sample point estimate alone does not. Such a rejection does not identify a unique microscopic mechanism, and it does not show that the residual survives as $D \rightarrow \infty$.

The registered scalar direction is tested in five primary cases. We use the one-sided Student- t reference with 11 degrees of freedom for the twelve panel pseudovalues and control the family error at 0.05 by a Bonferroni threshold of 0.01 per case. This finite-sample gate, rather than the normal approximation stored with the earlier raw-moment calculation, determines whether a cell enters the claim table.

C. Inference units and claim levels

Every uncertainty statement names its declared statistical unit. The appropriate unit may be an independent disorder realization, an independently drawn Hamiltonian, or a fixed-Hamiltonian tangent panel whose construction is treated as exchangeable under a frozen protocol conditional on the common base Hamiltonian. These cases answer different questions. Eigenvalues and matrix elements drawn from the same response matrix are not independent samples, and tangent directions evaluated at one fixed projector do not estimate Hamiltonian-to-Hamiltonian fluctuations. Pooling such quantities would understate uncertainty without adding an independent physical realization.

For fixed-base panels, products entering the Wick tensor are estimated with distinct panels, giving a U-statistic rather than a same-panel plug-in product. Uncertainty is obtained from delete-one-panel pseudovalues, as detailed in the Supplemental Material. For disorder or

independent-Hamiltonian ensembles, the same construction is applied with the full realization as the deleted unit. Comparisons between models retain their native units unless a matched sampling protocol has been carried out.

We consequently distinguish three levels of evidence. Curvature level repulsion is a finite-size statement about local eigenvalue correlations. Agreement of T_{abcd} with the complete W_{abcd} tests fourth-order closure for a specified tangent ensemble. Geometric ETH in an asymptotic sense would require a growing-rank sequence, a fixed operator class, and controlled decay of all covariance-whitened connected cumulants. The calculations in this paper are reported at the first two levels only where their respective gates have actually been evaluated.

IV. EXACT-DEGENERACY MECHANISMS AND MODEL FAMILIES

The models are grouped by the algebra that fixes the protected rank, rather than by spatial dimension or by the statistic that is eventually measured. This organization is necessary because the same dimension D can arise from a common kernel, a cohomology group, or dependent stabilizer constraints, and these mechanisms impose different restrictions on the response matrices. In every numerical case we identify the kernel first, verify a positive gap to its complement, and only then construct tangent responses. The ranks and gap audits used in the comparison are imported from the generated evidence table, not transcribed into the prose.

A. Constraint kernels and clustered Hall parents

A frustration-free parent can be written in terms of a constraint map C as

$$H_{\text{par}} = C^\dagger C = \sum_{\alpha} C_{\alpha}^\dagger C_{\alpha}, \quad \ker H_{\text{par}} = \bigcap_{\alpha} \ker C_{\alpha}. \quad (9)$$

Positivity pins every state in this common kernel to exactly zero energy. It does not fix the orientation of the kernel when the constraints change. If $C(\lambda)$ has constant nullity and its smallest nonzero singular value remains positive, Eq. (9) supplies precisely the constant-rank, gapped projector family assumed in Sec. II.

Two-body clustering and the Kapit–Mueller realization. The Laughlin state supplies the two-body clustering mechanism [12]. In a lowest Landau level or a Chern band, the bosonic V_0 pseudopotential may be factorized as

$$H_L = V_0 \sum_R A_R^\dagger A_R, \quad A_R |\Psi_0\rangle = 0, \quad (10)$$

where A_R annihilates a pair in the forbidden relative-angular-momentum channel and R labels its center-of-mass translate. Quasihole zero modes are common null vectors of all A_R ; their admissible-root counting is fixed by

the $(1, 2)$ clustering rule [13], and the zero-mode property admits a purely second-quantized formulation [14].

Our lattice realization uses N bosons on a square torus with n_ϕ orbitals in the exactly flat lowest Kapit–Mueller Chern band [15]. The Hilbert space is the fixed-particle-number bosonic Fock space after projection to that band. The boundary conditions are magnetic-periodic; boundary twists are introduced only when the closed twist torus is explicitly studied. Projecting the onsite repulsion gives the positive two-body parent in Eq. (10). The protected rank is the generated quasihole count in Table I, and a case is retained only when its first positive many-body eigenvalue gives a positive gap over the entire registered parameter set.

We use two logically distinct tangent classes. The first is *exactness-preserving constraint/generator transport*. A smooth one-particle unitary $U(\lambda)$, lifted to the fixed- N Fock space as $\Gamma_N[U]$, defines $C(\lambda) = C_0 \Gamma_N[U(\lambda)]^\dagger$ and $H(\lambda) = C(\lambda)^\dagger C(\lambda)$. Consequently $S_a = 0$ along the path, the nullity is constant, and the transported kernel remains exactly degenerate. Statements about finite-path parallel transport, holonomy, and protection mixing belong to this class.

The second class comprises the *v_2/v_3 Laughlin fixed-parent local-potential panels*. At the exact parent, structured Fourier and random local potentials enter through $Q(\partial_a H)P$. They define the linear response of the isolated rank- D cluster at the exact parent, but generally do not preserve exact degeneracy away from the base point. Curvature spectra and four-channel statistics obtained from these panels are therefore base-point response statements, not claims about a finite exact path. An intrafiber intervention is used only as the fixed-projector spectral control. Together, the two classes make the Laughlin parent a spatially local, large-rank comparison between exact transport and stochastic projector response protected at the base point by two-body clustering.

A term with a nonscalar $P\delta H P$ breaks Laughlin exactness. Concrete examples are residual band dispersion, a generic projected potential $\sum_{\mathbf{x}} \mu_{\mathbf{x}} n_{\mathbf{x}}$, or an interaction component outside the common V_0 kernel. Such terms need not destroy the surrounding fractional Hall phase, but at finite size they remove the exact zero-energy equality that is required here. Truncating the ideal Kapit–Mueller hopping can likewise replace the exact flat band by a narrow band; this distinction concerns exact degeneracy, not phase stability.

Three-body clustering and Moore–Read zero modes. The bosonic Moore–Read parent changes the order of the local constraint [16]. We work in the fixed- N bosonic Fock space of N_ϕ continuum lowest-Landau-level orbitals on a square torus. On a complete magnetic-translation orbit of coherent states, the factorized parent is

$$H_{\text{MR}} = \sum_{\mathbf{x}} C_{\mathbf{x}}^\dagger C_{\mathbf{x}}, \quad C_{\mathbf{x}} = \frac{1}{\sqrt{3!}} \left(\sum_j \phi_{xj} b_j \right)^3. \quad (11)$$

Every zero mode vanishes when three particles coincide. The fixed-four- quasihole sequence uses $N_\phi = N + 2$ with periodic boundary conditions. Its protected rank is obtained from the cyclic $(2, 2)$ admissible count and checked against the numerical nullity; the corresponding generated ranks and positive gaps appear in Table I. The factorized second-quantized construction is also the basis for the sparse numerical representation [17].

The tangent class consists of normalized guiding-center generators organized into local panels. Each Hermitian one-body generator G_a differentiates an invertible transport of the three-body constraint map. Write $\Gamma_a = \Gamma_N(G_a)$. On the protected fiber, the resulting tangent and response obey

$$(\partial_a H)P = -H\Gamma_a P, \quad X_a = Q\Gamma_a P, \quad S_a = 0. \quad (12)$$

Thus the v9 Moore–Read guiding-center panels are exactness-preserving generator transports, even though their response is evaluated analytically at the registered parent. Their curvature and four-channel statistics describe tangents to an exactly degenerate parent family, rather than generic splitting probes. This model’s role in the comparison is to change the clustering order and the zero-mode structure without introducing a supercharge or a stabilizer algebra. A generic two-body pseudopotential $\delta H_2 = \sum_R u_R A_R^\dagger A_R$, or any other term with nonscalar $P\delta H_2 P$, breaks Moore–Read exactness. Landau-level mixing and an incomplete three-body constraint orbit have the same effect at the level of the ideal parent problem.

B. Cohomological kernels

Random-supercharge cohomology. The random-supercharge model is zero dimensional: its Hilbert space is the fermionic Fock space of N complex modes, resolved into fixed-charge sectors. There is no spatial boundary condition. For a generic complex three-form C_{ijk} , define

$$Q(C) = \sum_{i < j < k} C_{ijk} \psi_i \psi_j \psi_k, \quad Q^2 = 0, \quad (13)$$

$$H_{\text{SUSY}} = \{Q, Q^\dagger\}.$$

This is the complex $\mathcal{N} = 2$ SYK construction [18]. Positivity implies that the protected fiber in a charge sector is $\ker Q \cap \ker Q^\dagger$, equivalently the space of harmonic representatives of Q -cohomology. Its protected rank is calculated from the charge-resolved cohomology and verified by exact nullity in every checkpoint. Because the opened calculation contains two charge sectors and many coupling realizations rather than one common D sequence, the v13 cross-model table leaves its single “sampled D ” cell as not tested instead of selecting a representative rank after the fact. The positive gap is the smallest nonzero Hodge-Laplacian eigenvalue in the same charge sector and is audited separately for every realization.

The tangent class consists of sparse and isotropic variations δC of the cubic supercharge. Differentiating H

separates the response into the orthogonal exact and coexact images of Q and $Q^\dagger$. The role in the comparison is a dense, nonlocal cohomological mechanism with an ensemble of independent coupling realizations. Deforming Q within the nilpotent cubic family preserves nonnegative energy and normally preserves the generic cohomology rank, although the rank may jump at singular coupling loci. By contrast, adding a generic charge-preserving interaction W that is not part of an anticommutator and has nonscalar PWP breaks random-supercharge exactness by splitting the harmonic multiplet.

Independence-complex cohomology on a lattice. For a graph G , let $d_i^\dagger = c_i^\dagger \prod_{j \sim i} (1 - n_j)$ create a fermion only when its neighboring vertices are empty. The allowed Hilbert space at charge f is therefore the vector space spanned by size- f independent sets. With complex site couplings z_i ,

$$Q = \sum_i z_i d_i^\dagger, \quad Q^2 = 0, \quad H_{\text{SUSY}} = \{Q, Q^\dagger\}. \quad (14)$$

This is the local lattice- $\mathcal{N} = 2$ construction [19]; its zero modes are computed by independence-complex cohomology [20].

The controlled family used here is $G = \bigsqcup_{s=1}^m C_6$, a disjoint union of six-site rings, in charge sector $f = 2m$. Each component has periodic boundary conditions; different rings have no connecting edge. The protected rank is 2^m and is checked directly against the zero-eigenvalue count, while the first positive eigenvalue supplies the gap gate. Local tangent panels vary the site couplings after projecting out the component-wise rescaling directions; isotropic panels provide a matched control. Both panels retain the exact/coexact Hodge decomposition. The role in the comparison is an exactly auditable local cohomology sequence, not a connected-lattice thermodynamic limit.

An unconstrained hopping term $\delta H_t = \sum_{\langle ij \rangle} t_{ij} c_i^\dagger c_j + \text{H.c.}$ or a nonuniform onsite potential generally cannot be written as the same Hodge Laplacian. When its restriction to the harmonic fiber is nonscalar it breaks lattice-SUSY exactness. Altering the exclusion projector so that the proposed supercharge is no longer nilpotent removes the cohomological protection more directly.

C. Commuting stabilizers

The X-cube model places one qubit on every edge of an $L \times L \times L$ cubic three-torus. Its Hilbert space has $3L^3$ qubits and periodic boundary conditions in all three directions. With cube X stabilizers A_c and vertex-plane Z stabilizers $B_v^{\mu\nu}$, the Hamiltonian is

$$H_{\text{XC}} = - \sum_c J_c A_c - \sum_{v, \mu\nu} K_v^{\mu\nu} B_v^{\mu\nu}, \quad J_c, K_v^{\mu\nu} > 0. \quad (15)$$

All stabilizers commute [21]. Their dependent relations on the torus leave $6L - 3$ logical qubits, hence protected

rank 2^{6L-3} . Positive coefficients retain the same simultaneous $+1$ eigenspace and a positive gap. Coefficient reweighting is therefore a fixed-projector tangent class. A separate local unitary conjugation transports the projector while preserving the entire spectrum; it supplies the scalar-curvature control. The role in the comparison is an exact structured counterexample rather than a stochastic ensemble.

Setting a stabilizer coefficient through zero closes the corresponding gap and changes the constraint count. Adding a logical membrane operator $-h\bar{Z}$ acts nontrivially within the ground space and breaks X-cube exactness immediately; a generic local Pauli field produces the same finite-size splitting only at the perturbative order needed to implement a logical operator. This makes the distinction between a protected phase and exact finite-size degeneracy explicit.

D. Quasi-degenerate checkerboard replication

The checkerboard calculation uses bosons in the projected lower band of a short-range, two-sublattice $C = 1$ hopping model on a periodic torus. Boundary twists enter the physical hopping matrix as seam phases before the band projection. The hopping amplitudes realize a nearly flat isolated band [22], and interactions produce an FCI-like low-energy multiplet [23]. Bond-magnitude, bond-phase, and site-potential disorder define the tangent path, and the onsite contact interaction is projected into the deformed lower band at each parameter point. The selected low-energy cluster has a nonzero internal bandwidth and remains separated from higher states by a positive external gap.

Accordingly, checkerboard FCI is a quasi-degenerate replication of the local geometric-statistics trend, not an exact-zero-mode mechanism. Its protected rank means the dimension of the tracked isolated cluster, not the nullity of a frustration-free parent. It tests sensitivity to short-range Chern-band microscopics; it is not included when statements require exact spectral silence.

V. SPECTRAL SILENCE AND GEOMETRIC CORRELATIONS IN LAUGHLIN MANIFOLDS

A. A degenerate spectrum with a resolved projector response

We begin with the obstruction that motivates the geometric calculation. In the registered spectral-silence instance, the zero-mode projector has rank $D = 50$. Its numerical bandwidth is $5.7825546 \times 10^{-16}$, whereas the first state outside the fiber lies at the positive gap 0.10667223 . The former is a residual audit and the latter is the scale that permits a reduced resolvent; neither is obtained by unfolding a narrow band. Within the accepted fiber all

energies are exactly equal at the algebraic parent, up to the reported solver residual.

The energy form factor consequently contains no information about the orientation of P . With the normalization used in Fig. 2, the raw contribution is the constant rank D , and the connected energy spectral form factor vanishes. This remains true for every tangent panel because the energy calculation uses only the repeated eigenvalue. It does not imply a vanishing response: the same projector has nonzero $X_a = Q(\partial_a P)P$ under the registered constraint or local-potential tangents.

Figure 2(b) makes this separation visible without invoking a single scalar summary. A structured curvature, a physical curvature panel, and a finite- D Jacobi reference can have the same matrix rank while having clearly different eigenvalue configurations. The physical spectrum fills the allowed interval and shows local repulsion, whereas the structured control occupies a much smaller set. Rank and exact energy degeneracy are therefore insufficient to determine the curvature point process.

The curvature form factor provides a second comparison. Over the registered finite- D window, the physical curve follows the Jacobi reference ramp within the uncertainty obtained from complete reference draws. At longer scales the difference need not remain small. Figure 2(d) and Fig. 3(d) show the surviving form-factor and number-variance residuals. We describe this as finite-rank ramp agreement with long-range memory. The statement is tied to the displayed rank, normalization, and tangent ensemble; it is not an asymptotic or universal claim.

B. The order in which correlations appear

To separate local and mesoscopic scales, the registered interpolation varies the strength g of the geometric scrambling while leaving the comparison protocol fixed. Figure 3(a) plots the curvature-form-factor residual over Fourier scale and g . The residual becomes small first in a restricted interval of Fourier scales. This flow is not uniform: a local spacing criterion becomes compatible with the finite- D reference before the first registered ramp window appears, as summarized in Fig. 3(b).

The ordering is physically useful. Nearest-neighbor repulsion tests only the shortest correlations of the curvature eigenvalues. A ramp requires correlations across a wider part of the point process, and number variance integrates correlations over a still longer window. Agreement of the first statistic thus need not force agreement of the other two. In the physical panel, the registered ramp residual becomes compatible with the finite- D reference over the shaded window in Fig. 3(c), while the number-variance difference grows beyond the marked local scale in panel (d).

The uncertainty unit changes with neither Fourier time nor window length. A complete physical orbit or a complete ensemble draw is one realization; the many eigenvalues and time samples within it are correlated observations.

TABLE I. Mechanisms that protect exact degeneracy and the evidence currently available for geometric statistics. The centered three-pairing statistic is denoted R_4^{full} . Its primary gates use exact registered finite-ensemble means, a Student- t_{11} finite-sample reference, and Bonferroni control over five cases. A missing calculation is written as not tested, not as a null result.

Model	Degeneracy mechanism	Geometric role	Sampled D	R_4^{full} primary split	Independent ensemble	Run state
Laughlin parents	two-body FQH parent	finite-size stochastic geometry	16, 25, 36	not tested	not tested	complete
$\mathcal{N} = 2$ SYK	random supercharge cohomology	finite-size stochastic geometry	not tested	not tested	not tested	complete
Moore–Read parent	three-body clustered FQH parent	finite-size stochastic geometry	42, 120	$N=4$: not passed $N=6$: passed finite rank	not tested	pilot only
Lattice $\mathcal{N} = 2$ SUSY	independence complex cohomology	finite-size structured geometry	2, 4, 8	$m=1,2,3$: not passed	not tested	pilot only
X-cube control	commuting projector subsystem code	exact structured control	not tested	not applicable	not applicable	complete

In particular, treating every curvature eigenvalue as an independent sample would artificially narrow the bands in Figs. 2 and 3. The complete physical panel is the statistical unit throughout this local-to-global comparison.

This hierarchy supplies the precise sense in which the geometry is Jacobi-like. It refers to local repulsion and a registered finite-rank ramp, not to equality of the entire joint distribution with an invariant ensemble. The long-range residual is evidence that the response retains information about its physical construction after the local spectrum has become compatible with the reference.

C. Intrafiber and off-fiber interventions

The separation between energy and geometry can also be imposed rather than inferred. Figure 4 compares two interventions on the same block decomposition. The intrafiber interpolation changes PHP . Its level statistics approach the registered GUE interval, but the projector is held fixed. The norm $\|P_\alpha - P_0\|_F$ and the curvature-spectrum error remain at numerical zero, so this operation changes the spectral channel without moving the quantum geometry.

The complementary interpolation changes the off-fiber block $P(\partial H)Q + \text{H.c.}$ while leaving the intrafiber intervention absent. The curvature gap statistic rapidly enters its Haar–Jacobi reference interval. Thus intrafiber spectral scrambling and off-fiber projector motion are independent response channels. Here *independent* means independently controllable in the block intervention; it does not mean that estimators measured from one physical panel are independent random variables.

The four quadrants in Fig. 4(d) are therefore all logically possible: neither channel may be random-matrix-like, either one can be compatible while the other remains structured, or both can be compatible. An exactly degen-

erate parent begins with a silent intrafiber spectrum, but it is not forced into the geometrically structured quadrant. This constructed control also prevents us from interpreting agreement in the curvature spectrum as indirect evidence that an unmeasured energy spectrum has become chaotic.

D. Historical Wick residual and the complete-cumulant boundary

We next turn from curvature eigenvalues to response matrix elements. The Kapit–Mueller calculations use particle numbers $N = 3, 4, 5$, with the audited quantities collected here to expose the finite-size range: The corresponding Fock-space dimensions are 120, 715, and 4368. Kernel count, gap, gauge invariance, leakage, and resolvent gates pass for each row before the matrix statistic is formed.

The statistic in Table II and Figs. 5(a,b) is historical. Its Gaussian null assumes a separable covariance and a proper complex response, for which the pseudocovariance pairing is set to zero. It consequently subtracts two Wick pairings. The physical local-density panels and the structured Fourier panel lie above that restricted null over

TABLE II. Kapit–Mueller zero-mode ranks, external gaps, and the historical four-channel residual. Every number is supplied by the generated v13 registry macros. The last two columns refer to the restricted separable proper-complex two-pairing null, not to the corrected complete cumulant.

N	D	Δ	$R_4^{2\text{pair}}$	$R_4^{2\text{pair}} - R_{4,\text{Wick}}^{2\text{pair}}$
3	16	0.055871585	0.37092804	0.15385124
4	25	0.10350202	0.24715444	0.1007779
5	36	0.093592675	0.20905928	0.081827266

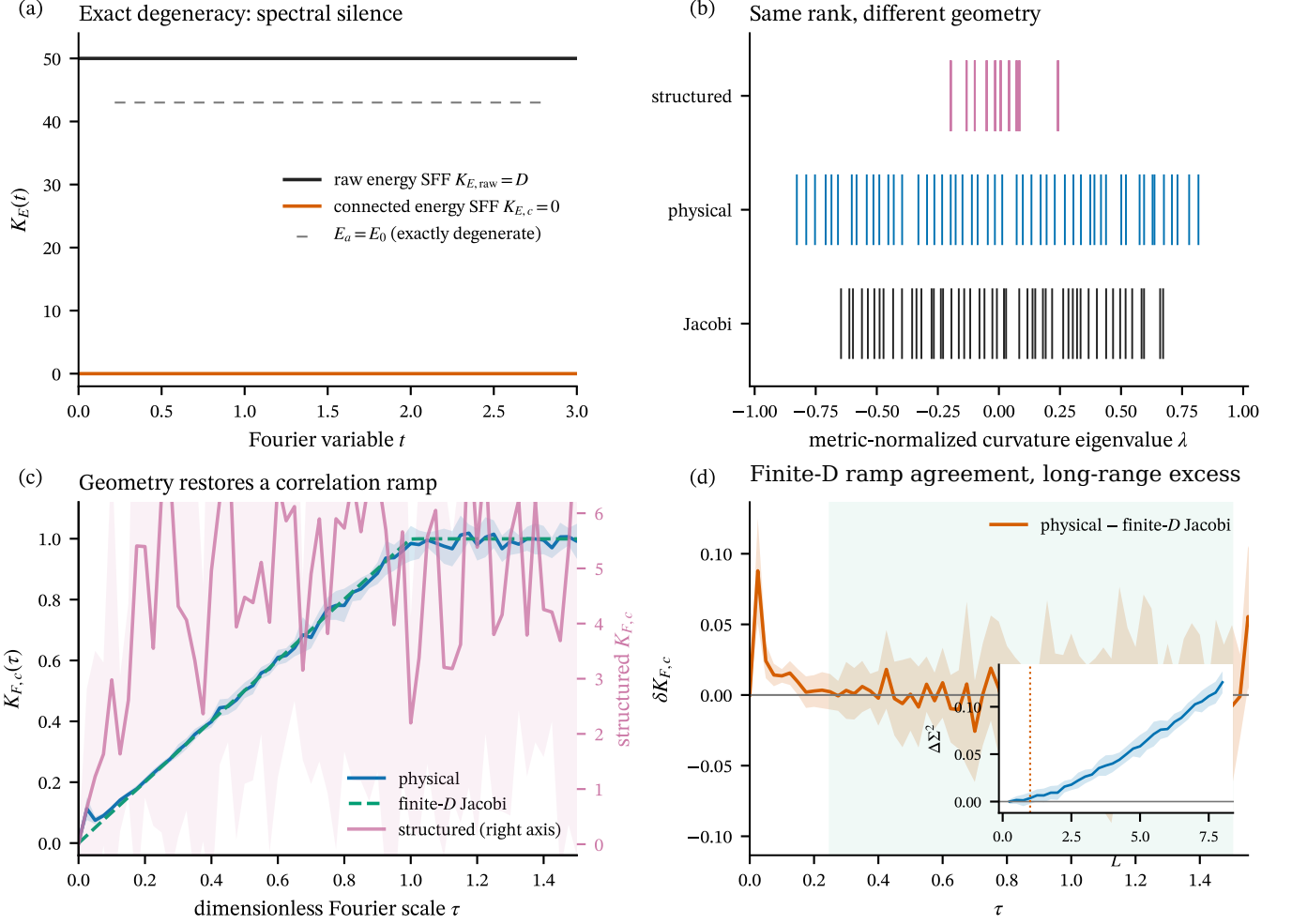


FIG. 2. Spectral silence and resolved projector geometry. (a) For the rank-50 zero-mode manifold, whose audited bandwidth and external gap are $5.7825546 \times 10^{-16}$ and 0.10667223 , the raw energy form factor is constant and its connected part vanishes. (b) Metric-normalized curvature spectra for a structured panel, a physical panel, and the finite- D Jacobi reference at the same rank. (c) The physical curvature form factor follows the registered finite- D ramp, whereas the structured control does not. (d) Physical-minus-reference residual; the inset shows the accumulated number-variance excess. Shading denotes uncertainty from complete physical panels or registered finite-Jacobi draws, not independent curvature eigenvalues.

the registered sizes. Their residuals decrease with N , as do the generated excesses in the last column.

This is a valid rejection of the stated restricted null, but it is not the corrected complete cumulant of Eq. (8). A positive historical two-pairing residual can be produced by nonseparable two-point covariance or nonzero pseudo-covariance without any connected four-point cumulant. We therefore do not call Figs. 5(a,b) a detection of non-Gaussian closure failure. A corrected claim would require the full entry covariance, all three complex Wick pairings, and uncertainty over declared independent units. The Laughlin row in Table I records that this complete-covariance gate has not been tested.

The physical error bars in Fig. 5(a) use 24 complete local-density panels. One entire panel, not one entry of X_a or one pair of indices, is resampled. The Fourier panel is shown as a structured comparison and is not pooled with

the local panels to manufacture an uncertainty estimate.

E. Parent dependence within two-body clustering

The continuum lowest-Landau-level calculation preserves the same two-body Laughlin constraint mechanism but changes the microscopic orbitals and the realization of the physical tangents. Figure 5(c) displays guiding-density, magnetic-bond, and magnetic-current panels separately. For the size-indexed registry branch, the exported connected excesses are 0.19378584, 0.14929835, and 0.10502783. The figure, rather than those three representative macros, resolves the full operator-class dependence and its complete-panel bootstrap interval.

The comparison between the Kapit-Mueller and continuum LLL parents is deliberately not a pooled test.

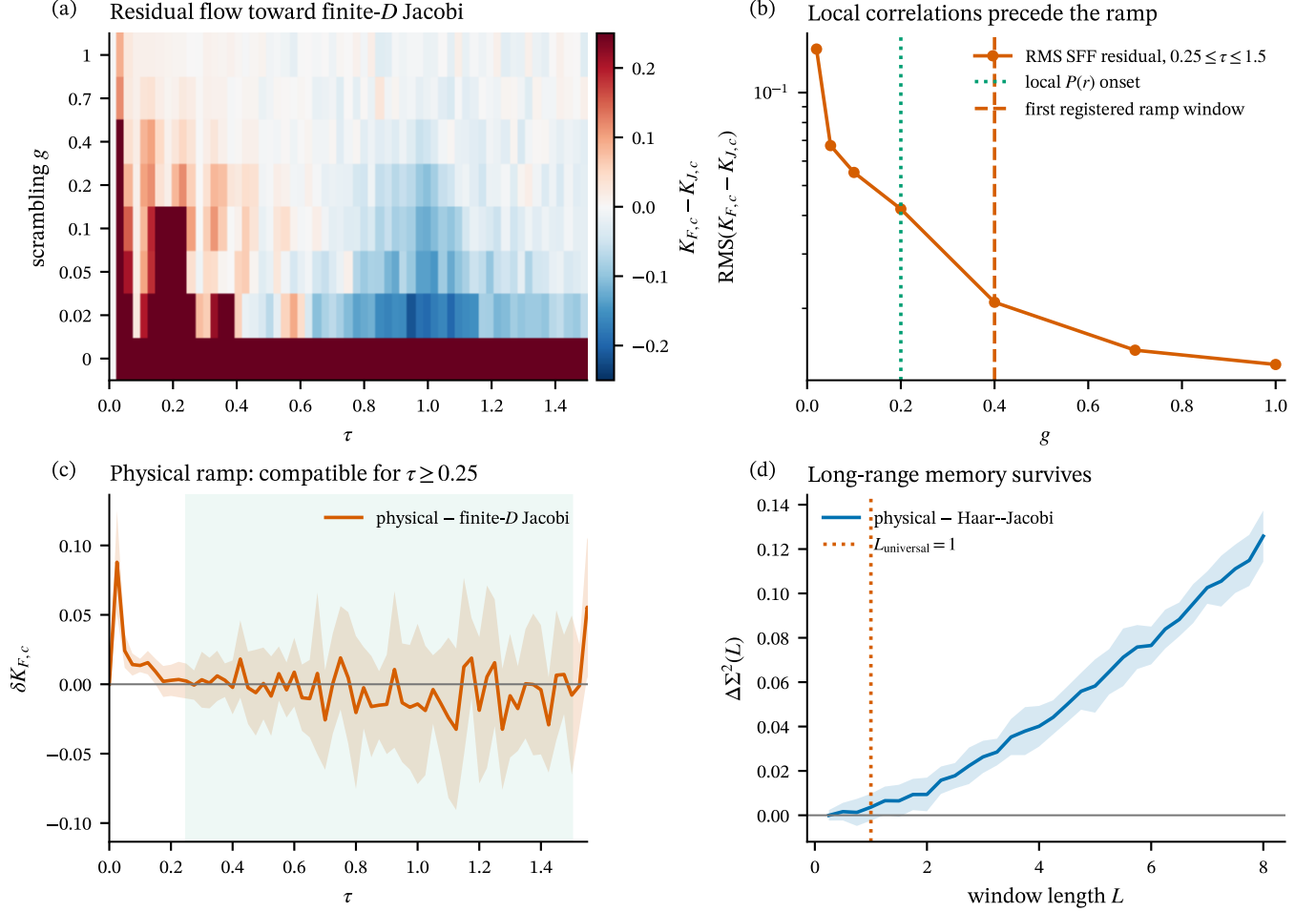


FIG. 3. Local-to-global hierarchy of curvature correlations. (a) Difference between the physical and finite- D Jacobi connected form factors versus Fourier scale τ and geometric-scrambling strength g . (b) The local spacing-ratio criterion becomes compatible before the first registered ramp window. (c) In that window the physical ramp residual is consistent with the finite-rank reference. (d) The connected number-variance difference grows beyond the marked local scale, retaining long-range memory. Curves and bands use a complete orbit or ensemble draw as the uncertainty unit; repeated τ , window-length, and eigenvalue samples within one draw are not treated as independent.

Both possess a positive factorized V_0 parent and the same exact constraint-transport identity, but their complement spectra and local generator maps differ. Agreement would have supported a parent-insensitive scalar law within this protection mechanism; the registered outcome instead selects parent dependence.

Specifically, the sealed common-plateau prediction was evaluated without refitting on the three continuum tangent classes and three registered sizes. Figure 5(d) reports that zero of nine registered predictions cover their corresponding observed points. The error bars there are complete-panel observation-bootstrap intervals; the registered prediction is not adjusted after seeing the continuum outcomes.

The no-refit failure has a narrow meaning. It rejects that particular scalar law as a simultaneous description of the displayed operator classes and sizes. It does not imply the absence of finite-size decay: all three continuum classes

decrease over the shown range. It also does not show that no refitted function could describe the data, that the two microscopic parents occupy different phases, or that two-body clustering lacks a common local curvature regime. The result is parent and operator dependence of the tested four-channel observable at finite rank. It supplies no universality or asymptotic conclusion.

VI. GEOMETRY AND TOPOLOGY AT FIXED CHERN CLASS

The local curvature statistics studied above do not determine the topology of the protected bundle, and the converse is also false. To separate these levels, we transport the Laughlin constraint map over the boundary-twist torus (θ_x, θ_y) while applying a periodic ambient unitary deformation of strength g . At every twist and

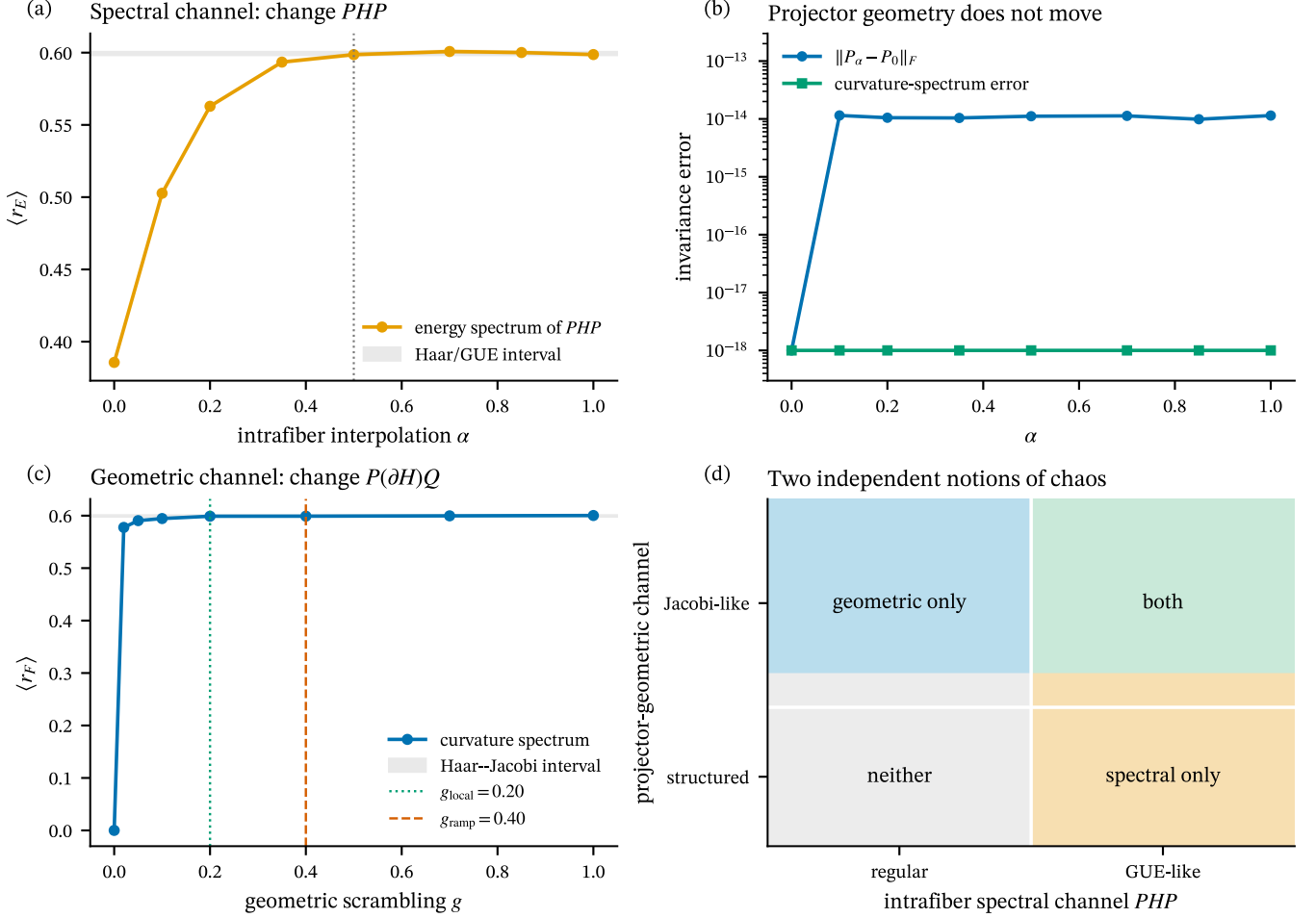


FIG. 4. Independently controllable spectral and geometric channels. (a) An intrafiber interpolation changes PHP and brings its gap statistic to the registered Haar/GUE interval. (b) The projector and curvature spectrum remain fixed to numerical precision during that intervention. (c) A separate off-fiber interpolation changes $P(\partial H)Q + \text{H.c.}$ and brings the curvature statistic to its Haar–Jacobi interval without using the intrafiber interpolation. (d) The two axes permit structured, spectral-only, geometric-only, and jointly compatible regimes. Uncertainty bands are formed from the registered interpolation-endpoint ensemble. *Independent* here means separately tunable response channels, not independent entries of one response matrix.

every accepted g ,

$$H_g(\theta) = \mathcal{V}_g(\theta) H_0(\theta) \mathcal{V}_g^\dagger(\theta).$$

This is an exactness-preserving path rather than a fixed-parent probe. The many-body spectrum is identical point by point on the twist torus, the protected rank stays fixed, and the external gap never closes.

A. Fixed rank, gap, and first Chern number

The two registered bundles have ranks 16 and 25. Their minimum gaps over the twist mesh and deformation family are respectively 0.051741368 and 0.094695221. In addition to checking the kernel residual at each vertex, we require the smallest singular value of every neighboring-frame overlap to exceed 0.85325072 for the first bundle and 0.83693205 for the second. These overlap floors exclude

a mesh on which the polar parallel transport becomes ill-conditioned.

The determinant line bundle is unchanged along the deformation. The discretized trace curvature gives first Chern numbers 6 and 10 for the two ranks, with the same integers on the primary and convergence meshes. Figure 6(a) shows these constants together with the open external gaps. The computation also compares the determinant Wilson-loop phase with the integrated trace curvature; their agreement is a gauge invariant check that the mesh winding and the plaquette Chern calculation refer to the same Abelian sector.

Panel (b) illustrates this constraint for the rank-25 bundle. The determinant phase flow changes shape between the endpoints, but its net winding remains 10. Thus topology fixes the integer winding, not its distribution along the transverse twist. The same distinction appears already at the level of local curvature: matrices can redis-

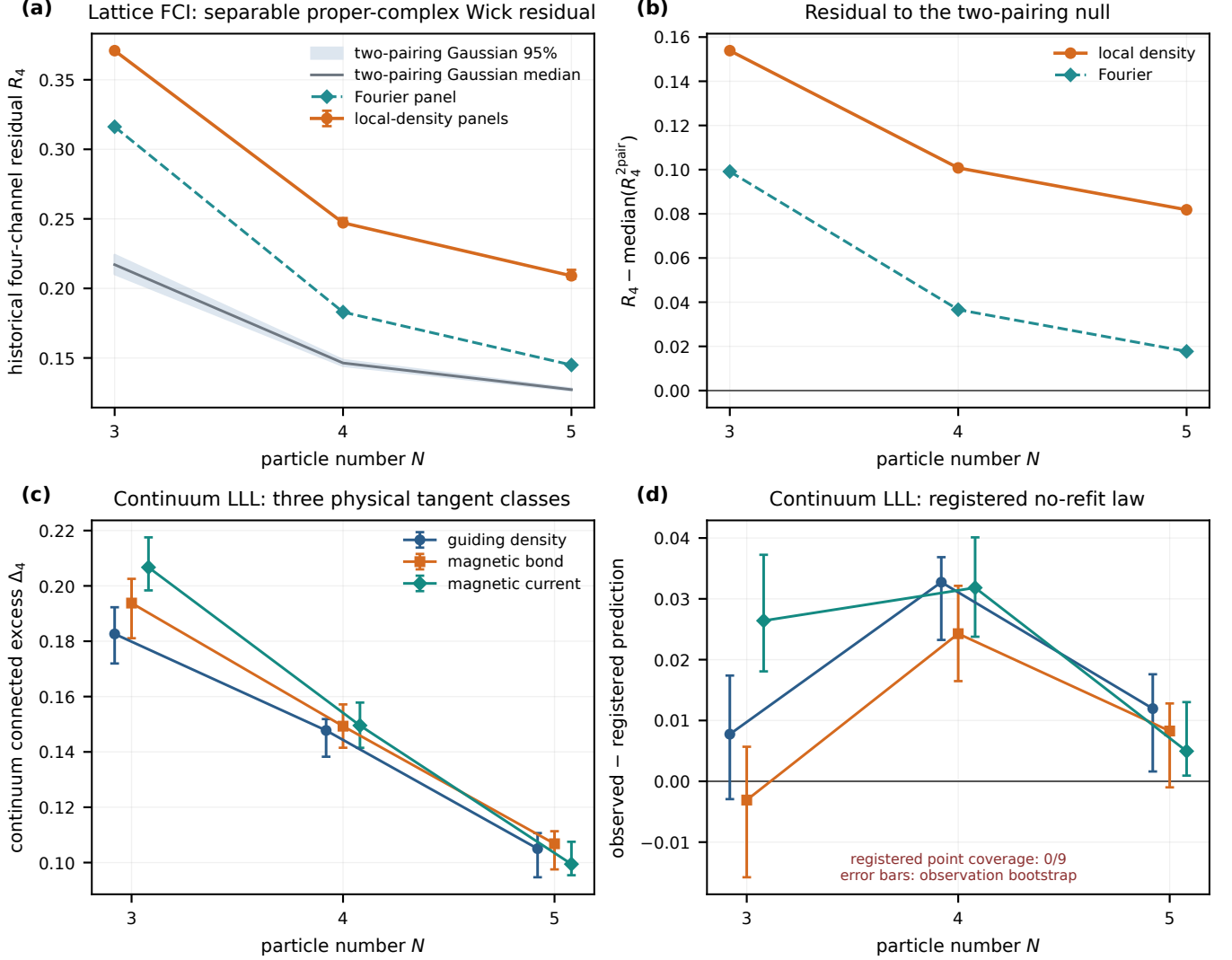


FIG. 5. Restricted Wick residuals and parent dependence. (a) Kapit–Mueller local-density and Fourier panels compared through the historical two-pairing residual under a separable proper-complex Gaussian null. (b) Residual relative to that two-pairing median. This is not the corrected complete cumulant, which requires the full covariance, pseudocovariance, and all three complex Wick pairings. (c) Continuum-LLL connected excess for guiding-density, magnetic-bond, and magnetic-current panels; error bars bootstrap complete physical panels. (d) Observed-minus-predicted residual for the sealed no-refit common-plateau law. The registered point coverage is zero of nine. This rejects that law across the displayed classes and sizes; it does not reject finite-size decay or all refitted forms. The two parents are not pooled into one uncertainty estimate.

tribute their traceless part while preserving the integral of $\text{Tr } F_{\theta_x \theta_y}$.

B. Non-Abelian holonomy inside a fixed topological sector

For each fixed transverse twist, the polar factors of adjacent frame overlaps are multiplied around the θ_x cycle to form a Wilson matrix $W_x(\theta_y)$. A frame change conjugates W_x , so its eigenphases, gap ratio, and normalized spectral form factor are gauge invariant. The determinant of W_x probes the Abelian winding discussed above; relative

eigenphases probe the internal $SU(D)$ holonomy.

The internal holonomy changes under noncommuting ambient generators even though the energy spectrum and Chern number do not. The median Wilson gap ratio moves from 0.28849403 to 0.32114901 in the rank-16 bundle and from 0.30997725 to 0.33175374 in the rank-25 bundle. The registered generator ensemble, rather than individual Wilson eigenphases, is the uncertainty unit in this comparison.

This change is statistically resolved relative to its generator-to-generator spread, but it does not carry the Wilson spectrum to the circular unitary ensemble (CUE). Figure 6(c) places the endpoint gap ratios below the CUE

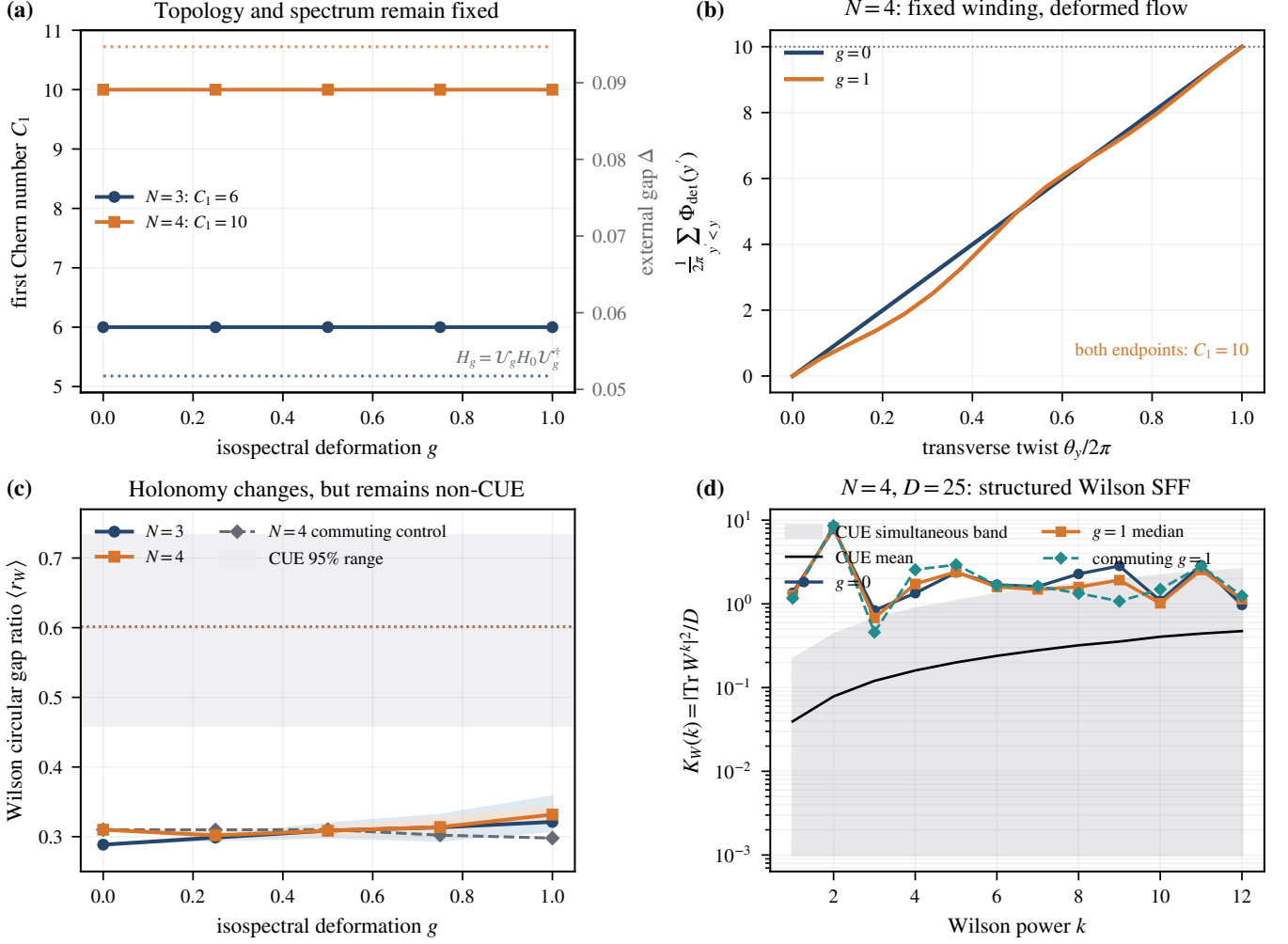


FIG. 6. Non-Abelian holonomy along an exactly isospectral, fixed-Chern transport. (a) The ranks, first Chern numbers, and positive external gaps remain fixed under ambient unitary deformation. (b) For $N = 4$, $D = 25$, the determinant Wilson-loop flow deforms while retaining winding 10. (c) The internal Wilson gap ratio changes for noncommuting generators, while the commuting control remains structured; neither endpoint enters the CUE reference band. (d) The normalized Wilson form factor likewise retains non-CUE peaks. Marker intervals use the registered generator ensemble, and the gray CUE uncertainty bands use independent circular-unitary reference draws. Topology and the exact energy spectrum are unchanged throughout.

reference band, and panel (d) shows that the Wilson form factor retains pronounced structured peaks. The largest registered deformation is therefore not CUE-compatible. The commuting-generator control remains near the undeformed curve, confirming that the observed change uses the noncommuting part of the transport rather than the Chern calculation or an uncontrolled change of spectrum.

The result has two immediate consequences. First, a fixed first Chern number does not fix the non-Abelian holonomy. Bundles with the same rank, energy spectrum, external gap, and Chern integer can have distinguishable internal Wilson spectra. Second, changing the holonomy within that sector does not by itself produce Haar statistics. Topological protection supplies the bundle on which the geometric observable is defined; it neither forbids internal mixing nor guarantees random-matrix closure.

The isospectral construction is important for this interpretation. Had the gap or protected rank changed with g , the holonomy shift could have been attributed to a changing spectral cluster. Had only a fixed-parent tangent been evaluated, there would be no finite exactly degenerate bundle on which to define the closed Wilson loop. Here the equality $H_g = \mathcal{V}_g H_0 \mathcal{V}_g^\dagger$ holds over the complete twist torus, so the difference between Abelian topology and non-Abelian geometry is measured within one exact transported family.

Gauge covariance is checked separately from mesh convergence. Random unitaries applied to every local frame change individual link matrices but leave the Chern integer and Wilson eigenphases invariant. Increasing the twist mesh reproduces the accepted integers and endpoint winding, while the overlap singular-value gate controls

the stability of the polar factors. These checks matter because a smooth-looking eigenphase flow can otherwise hide a branch-cut or coarse-mesh error. The reported holonomy change survives all three tests: random frame rotation, mesh refinement, and an open external gap.

These two finite-rank bundles do not define a large- D sequence. The Chern integers also differ between the two particle numbers, so they cannot be used as evidence for size-independent Wilson statistics at a single fixed topological charge. What is established is a controlled counterexample to the implication that a fixed Chern class fixes internal holonomy. Testing an asymptotic geometric law at fixed topology would require a growing-rank sequence with the same topological sector, tangent prescription, mesh convergence criteria, and independent generator ensemble.

VII. THREE-BODY CLUSTERING AND COHOMOLOGICAL RESPONSE

Figure 7 compares three protected constructions for which the response is nonzero: a three-body clustering kernel, a random supercharge, and a local lattice supercharge. The common feature is an exactly degenerate zero-energy fiber separated from its complement. The origin of that fiber and the sampling measure are different. We therefore compare the algebra and the statistical result within each model before placing their evidence cells next to one another. In particular, a tangent panel at one fixed Hamiltonian is not interchangeable with an independent disorder realization.

A. Moore–Read: an exact moving kernel

The Moore–Read parent is a sum of positive three-particle constraints, Eq. (11). The states in its kernel obey the same three-body clustering condition at exactly zero energy [16, 17]. For the fixed-four-quasihole sequence $N_\phi = N+2$, the two opened systems have $(N, D) = (4, 42)$ and $(6, 120)$. The first positive energies, 1.67285 and 1.5351551, respectively, verify that the response is taken for an isolated projector rather than for a numerically selected portion of a continuum.

The guiding-density directions are an exactness-preserving generator transport. If G_a is lifted to the N -particle Fock space by $\Gamma_N(G_a)$, differentiation of the transported constraint gives

$$(\partial_a H)P = -H\Gamma_N(G_a)P, \quad X_a = Q\Gamma_N(G_a)P. \quad (16)$$

Thus $S_a = P(\partial_a H)P = 0$, while the off-fiber block X_a is generally nonzero. This identity is checked against the parent action and complement leakage for every stored panel. It matters conceptually: the response is not the derivative of a multiplet that immediately splits. It is the derivative of an exactly degenerate kernel that moves under a finite invertible constraint path.

The finite-size calculations used two related, but inequivalent, fourth-order statistics. The historical two-pairing residual in Fig. 7(a) subtracts a separable proper-complex Gaussian prediction. Its local-panel medians are 0.50164711 and 0.30930182; the structured-line controls are 0.58584377 and 0.39008828. Those values describe the old $R_4^{2\text{pair}}$ observable. They neither include the pseudocovariance pairing nor reproduce an arbitrary entrywise covariance, so they are not a test of the complete Gaussian law.

The v12 raw, uncentered three-pairing calculation is also retained only for provenance. Its response arrays were neither population-centered nor channel-whitened before the moment subtraction. The corresponding raw estimates, 0.19087579 and 0.19360477, cannot be interpreted as connected cumulants. A nonzero raw value may contain the finite-population mean and is therefore claim-ineligible even when a nominal normal-reference probability is small. Figure 7(b) marks this distinction explicitly: “exact fiber and response” and “centered complete covariance” are separate evidence questions.

For the corrected R_4^{full} calculation, one fixed-base tangent panel is the statistical unit. Each system has a registered finite population of torus guiding-density anchors, and linearity gives its exact population mean before any panels are assigned to inference. The mean vanishes for the uniform traceless guiding-density population; it is not estimated from the observed inference half. The 12 panels in the training half determine only the 2×2 channel Gram matrix. Its inverse square root is then frozen and applied to the disjoint 12 panels. We refer to this as the frozen 12/12 split. The training half alone fixes the whitening transformation. All three Wick pairings, including the pseudocovariance term, are subtracted from the inference-panel U-statistic.

Uncertainty is obtained by deleting one complete inference panel and recomputing the contraction. The registered scalar direction is tested with a Student- t reference with 11 degrees of freedom. A Bonferroni correction controls the family level 0.05 across five registered primary cases (5 in total), giving per-case level 0.01. This protocol is conditional on exchangeability of the fixed-base panels; it is not an estimate over Moore–Read Hamiltonians or over parent-Hamiltonian families.

At $N = 4$, $D = 42$, the directional estimate is 0.092095827, while the familywise one-sided lower bound is -0.11524792. The registered positive-direction gate does not pass. At $N = 6$, $D = 120$, the corresponding estimate is 0.18490062 and the familywise lower bound is 0.039160542; this case passes. The positive cell in Fig. 7(c) is consequently a finite-rank $N = 6$ result. It is not an asymptotic statement, and the absence of a pass at $N = 4$ forbids a monotone scaling claim from these two ranks alone. The reverse split is descriptive and is not pooled with the primary split, even where its point estimate has the same sign.

This result establishes a connected fourth-order response contraction at one rank for the stated tangent

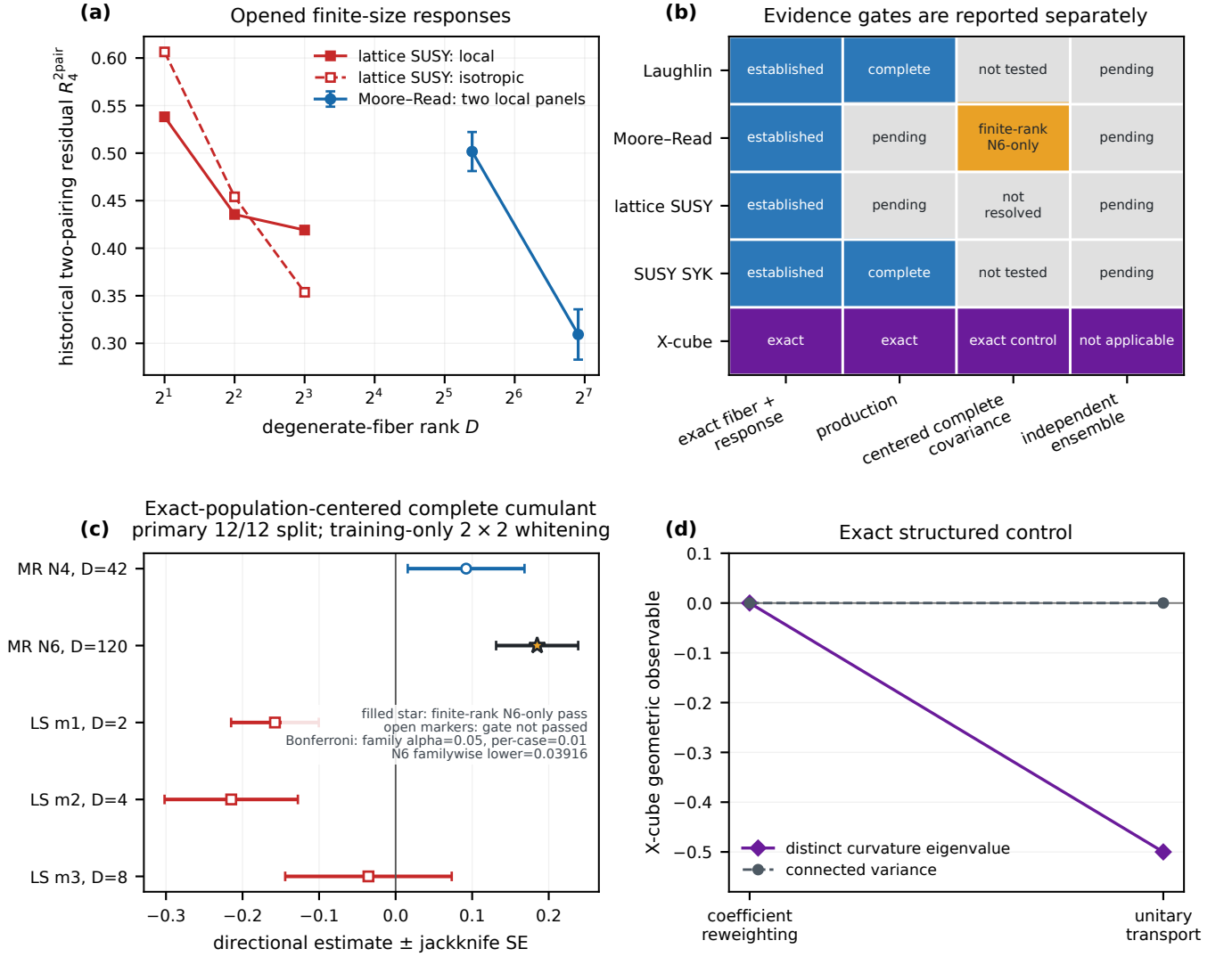


FIG. 7. Mechanism-resolved response under exact degeneracy. (a) historical two-pairing residual $R_4^{2\text{pair}}$ for local and isotropic lattice SUSY panels and two local Moore-Read panels. This statistic uses a separable proper-complex null and is not the corrected complete cumulant. (b) Evidence cells are kept separate: exact fiber and nonzero response, production status, centered complete covariance, and an independent model/operator ensemble are different gates. (c) Directional estimates of the exact-population-centered complete cumulant use a frozen 12/12 split, training-only 2×2 channel whitening, all three Wick pairings, and delete-one-panel errors. The filled star marks the sole familywise finite-rank pass, Moore-Read $N = 6$; open markers do not pass. The reverse split is descriptive and not pooled. (d) X-cube coefficient reweighting has zero curvature, whereas local unitary transport has deterministic scalar curvature and zero connected variance. These are exact structured controls, not Gaussian samples.

population after complete covariance and pseudocovariance subtraction. It does not establish that all fourth-order components are nonzero, that the effect persists with D , or that different three-body parents share the same law. It also does not identify a dynamical Lyapunov exponent: the measured object is the geometry of an exactly degenerate projector over coupling space.

There are two further reasons not to read the $N = 4, 6$ comparison as a fit. First, the ambient occupation-space dimensions and the protected ranks change together, so a change in the scalar contraction cannot be assigned

uniquely to D . Second, the response channels are generated by a discrete torus population whose geometry also changes with N_ϕ . An asymptotic test would have to hold the operator prescription fixed in physical units, open additional ranks, and preregister the scaling quantity before examining the new outcomes. Here the two cases instead test whether the same finite-panel protocol can be carried out at two nontrivial ranks and whether its familywise gate is resolved in either one.

The fixed-base qualification does not contradict exact transport. Every G_a defines a finite isospectral constraint

path, but the collection of panels is evaluated at the same base parent and base projector. The jackknife therefore samples the registered distribution of tangent directions through one point of the projector bundle. It does not sample different points along a path, and it does not absorb the correlation among channels within one panel into the sample count. This distinction is why the full response tensor of a panel, rather than each generator or matrix entry, is deleted in the uncertainty calculation.

B. Random-supercharge cohomology

The random supersymmetric model replaces a clustering ideal by a chain complex. A cubic nilpotent supercharge Q maps a charge- r sector to the charge- $(r-3)$ sector, and

$$H_r = Q_{r+3}Q_{r+3}^\dagger + Q_r^\dagger Q_r \quad (17)$$

is its Hodge Laplacian. Its zero modes are harmonic representatives of Q -cohomology, $\ker Q_r / \text{im } Q_{r+3}$, and hence are annihilated by both Q and $Q^\dagger$ [18]. Exact degeneracy follows from nilpotency and positivity, not from spatial locality. This is the finite supersymmetric sector in which non-Abelian Berry curvature was previously proposed as a chaos probe [9].

Let $\Pi_- = Q_{r+3}Q_{r+3}^\dagger H_r^+$ and $\Pi_+ = Q_r^\dagger Q_r H_r^+$ be the orthogonal projectors onto the exact and coexact positive-energy subspaces, where H_r^+ is the Moore–Penrose inverse of H_r on the complement of its kernel. Differentiating the two Hodge terms then separates the response into $X_a^- = \Pi_- X_a$ and $X_a^+ = \Pi_+ X_a$,

$$X_a = X_a^- \oplus X_a^+, \quad \langle X_a^-, X_b^+ \rangle_{\text{HS}} = 0. \quad (18)$$

Here the direct resolvent response and the sum of the two branch responses are checked independently. The branch weights, channel covariances, and left/right covariance spectra can therefore be used to build a Hodge-resolved Gaussian comparison without opening the fourth-order outcome. This branch structure is physical information imposed by the complex; it should not be discarded by replacing the full response with an isotropic rectangular matrix.

The registered $N = 14$ test used a complete independent disorder realization as the uncertainty unit. The adjacent- and central-charge sparse panels have observed median historical residuals 0.30152872 and 0.37499257. Before those outcomes were unsealed, the null distribution was generated from the stored two-point Hodge signatures. The held-out medians lie outside both the collapsed and Hodge-resolved prediction intervals. This rejects the frozen separable proper-complex Gaussian null for the registered response statistic.

The qualifier “separable” is essential. That null matches marginal channel, target, external, and exact/coexact covariance spectra, but it is not the complete entrywise

covariance C together with pseudocovariance Π . Consequently, its rejection can arise from a connected fourth cumulant or from nonseparable two-point structure omitted by the null. The random-SYK cell for centered R_4^{full} remains not tested. We do not use the historical residual to claim complete-covariance non-Gaussianity, and we do not pool its independent-realization bootstrap with the fixed-base panel jackknife used for Moore–Read.

This separation also identifies the useful next test. Random couplings supply genuine Hamiltonian-to-Hamiltonian sampling, whereas the Moore–Read calculation supplies an exact moving-kernel path at one parent. A common claim would require the same centered full-covariance observable and a matched tangent prescription in both ensembles. Those cells are absent from the present registry.

The Hodge split also changes the meaning of apparent isotropy. Unitary rotations within the exact image and within the coexact image may leave all stored covariance spectra invariant while changing correlations between the two embedded subspaces and the protected fiber. A collapsed null that keeps only the total left and right spectra forgets even more information. The sealed v7 comparison was designed to ask whether these progressively richer separable nulls predicted the held-out median. Their failure is reproducible evidence for covariance memory beyond those summaries; it is deliberately not relabeled as the complete cumulant of Sec. III.

C. Local lattice cohomology

The lattice construction retains the Hodge mechanism but changes both locality and sampling. On a union of m six-site cycles, the nilpotent supercharge acts on the independence complex, and its harmonic sector has ranks $D = 2, 4$, and 8 for $m = 1, 2, 3$. The corresponding positive gaps are 0.17454073, 0.30663782, and 0.14619751. Local changes of the site couplings move the harmonic projector. The response again splits into exact and coexact branches; their measured local-panel balance factors are 0.66096261, 0.50544553, and 0.24583911. Thus the local family is neither a fixed-projector control nor a single-branch rectangular ensemble.

Panel (a) shows the historical two-pairing residual for local and isotropic tangents. The local values 0.53819492, 0.43553404, and 0.41910381 differ from their isotropic counterparts 0.60644423, 0.45406292, and 0.35365135. These finite-size differences show that the accessible tangent class matters. As in Moore–Read, however, the old $R_4^{2\text{pair}}$ numbers do not answer the complete-covariance question. The v12 raw estimates are likewise uncentered and are not used for a claim.

The corrected calculation uses the exact population mean obtained by enumerating the registered ordered local-coordinate marginals. This mean is nontrivial because the discrete local-tangent construction is not symmetric entry by entry. After population centering, the

same frozen 12/12 split, training-only channel whitening, three-pairing U-statistic, Student- t reference, and five-case Bonferroni family are used. This alignment makes the Moore–Read and lattice rows comparable at the level of the estimator, but it does not turn the lattice panels into independent graph realizations.

Disjoint cycle components make the algebraic rank sequence exactly auditable, but they also impose a tensor-product ancestry on the cohomology. Increasing m simultaneously adds degrees of freedom and repeats a local building block. The sequence is therefore useful for checking how a known branch structure enters the response and the estimator; it is not a connected- lattice thermodynamic sequence. A claim of locality-driven universality would require connected graphs, a matched local tangent density, and an independent ensemble at each size. None of those missing conditions can be replaced by treating the many coordinate pairs at one graph as independent Hamiltonians.

The primary directional estimates are -0.15767575, -0.21481044, and -0.035495439. Their familywise lower bounds are -0.3133506, -0.45153645, and -0.3311967. None of the three registered positive-direction gates passes; failure of the positive-direction gate does not prove Gaussianity. It supplies neither an equivalence test nor a two-sided upper bound on every component of $\mathcal{K}$. A negative point estimate in a preselected direction may coexist with a nonzero cumulant in a different direction, and a small finite panel family may lack power. The reverse split is descriptive here as well.

The appropriate conclusion is therefore asymmetric. Local lattice cohomology supports an exact, moving, branch-reducible projector with nontrivial response, but the registered positive full-cumulant contraction is not resolved for $D = 2, 4, 8$. Moore–Read resolves that contraction only in the displayed rank-120 case. Random supercharge SYK rejects a narrower separable null using independent disorder realizations, while its complete entrywise-covariance cell remains open. These are three different statements, not three measurements of a universal number.

VIII. STABILIZER COUNTEREXAMPLES

The X-cube model provides an exact control in which the protected rank grows rapidly but the quantum geometry is fully determined by the tangent algebra. On the three-torus its dependent stabilizer relations leave $6L - 3$ logical qubits and ground-space dimension 2^{6L-3} [21]. Neither a large D nor a moving projector is therefore sufficient to produce random-matrix curvature. Figure 7(d) shows two deformations that separate these possibilities without allocating the exponentially large ground-space matrix.

The calculation is performed in the binary symplectic representation. Each Pauli stabilizer is a row $(\mathbf{x}|\mathbf{z})$ over $\mathbb{F}_2$, commutation is a symplectic inner product, and row reduction determines the number of independent

constraints. This proves the logical-qubit count without diagonalizing a 2^{3L^3} -dimensional Hamiltonian. It also exposes the relation responsible for the transported curvature: the ordered product of the chosen local generators is reduced against the stabilizer row space, rather than inferred from a small-system spectrum.

A. Positive coefficient reweighting

First vary the positive coefficients multiplying the commuting stabilizers while keeping the Pauli generators fixed. The simultaneous +1 eigenspace does not depend on those coefficients as long as none crosses zero. Hence the projector is parameter independent,

$$\partial_a P = 0, \quad X_a = 0, \quad g_{ab} = F_{ab} = 0. \quad (19)$$

The registered coefficient reweighting curvature is exactly 0. This is stronger than a small numerical residual: it follows from the common stabilizer eigenspace. The deformation changes excitation energies and may change the gap scale, yet it cannot rotate the protected subspace. Energy dependence outside the fiber and geometry of the fiber are independent data.

This control also identifies the relevant failure boundary. A coefficient passing through zero removes one constraint and closes the corresponding gap; a logical operator added within the ground space makes PHP nonscalar and splits the exact multiplet. Neither operation belongs to the open, constant-rank family of Eq. (19).

Generic local Pauli fields need not split the torus ground space at first order because a detectable error maps it out of the code space. Their finite-size splitting appears only at the perturbative order at which a noncontractible logical operator is assembled. That exponentially or high-order small splitting is physically important for code stability, but it is distinct from exact degeneracy at the commuting-projector point. Our controls stay on paths for which the algebra certifies exact equality of all ground energies.

B. Local isospectral unitary transport

The second deformation conjugates the full commuting-projector Hamiltonian by local unitaries. Let

$$H(\lambda_1, \lambda_2) = U H U^\dagger, \quad U = e^{-i\lambda_1 G_1} e^{-i\lambda_2 G_2}. \quad (20)$$

The spectrum, degeneracy, and gap are then preserved exactly, while $P(\boldsymbol{\lambda}) = U P U^\dagger$ moves. The registered local Pauli generators are individually detectable, anticommute, and have an ordered product equal to a stabilizer up to phase. Their commutator therefore acts centrally on the code space. The curvature is nonzero but scalar,

$$F_{12} = -0.5 P, \quad D^{-1} \text{Tr} \left[(F_{12} - D^{-1} \text{Tr}(F_{12}) P)^2 \right] = 0. \quad (21)$$

The dense low-rank toy audit fixes the sign in Eq. (21); the production statement follows from binary symplectic identities for every opened size. Thus unitary transport is exactly isospectral and geometrically nontrivial, but its non-Abelian curvature contains no fluctuating eigenvalue sector. Parallel transport around an infinitesimal loop gives a common phase rather than relative mixing within the code space.

Because the two generators anticommute, their ordered infinitesimal transports do not cancel. Because their product is a stabilizer, however, the resulting commutator has the same action on every code state. This is a concrete example of noncommuting motion in the ambient Hilbert space whose projection to the fiber lies in the center. It rules out the converse shortcut that any noncommuting deformation must generate a fluctuating non-Abelian spectrum.

The scalar curvature is compatible with a nonzero metric: the projector can move in the ambient Hilbert space even though the commutator of two accessible motions becomes central after restriction to the fiber. What vanishes is the traceless curvature component used for level-repulsion and connected-variance tests. This distinction prevents a nonzero Berry phase from being mistaken for stochastic non-Abelian mixing.

The connected variance vanishes because the response algebra collapses to the center on the protected fiber. There is no ensemble of fluctuating curvature matrices to which a Jacobi or Gaussian comparison should be applied. The complete-covariance gate is not applicable to this exact control; labeling it “passed” or “failed” would confuse deterministic centrality with a statistical null.

Together, the two X-cube paths exclude two shortcuts in interpreting the other models. The flat path shows that exact degeneracy can coexist with no projector response. The unitary transport path shows that nonzero response and curvature can coexist with zero connected variance. Thus large exact degeneracy is not sufficient for geometric chaos; one must also establish a noncentral accessible response algebra and then test its statistics with an appropriate sampling measure.

IX. PROTECTION, MIXING, AND STATISTICAL CLASSES

The examples above suggest a classification with three logically independent entries. The *protection class* records why a rank- D multiplet is exactly degenerate. The *mixing class* records which directions of the Grassmannian are reached by the allowed Hamiltonian tangents. The *statistical class* records the measure under which a statement about those directions is made. Keeping these columns separate is necessary: protection does not determine mixing, and mixing at one base Hamiltonian does not determine Hamiltonian-to-Hamiltonian statistics.

A. Response algebra and commutant

At a fixed protected projector define the accessible off-fiber space and its fiber algebra by

$$\begin{aligned}\mathfrak{X} &= \text{span}\{X_a\} \subset \text{Hom}(P\mathcal{H}, Q\mathcal{H}), \\ \mathfrak{A}_X &= \text{alg}\{X_a^\dagger X_b\}_{a,b} \subset \text{End}(P\mathcal{H}).\end{aligned}\quad (22)$$

The response rank is the dimension of $\mathfrak{X}$ after exact tangent-null directions are removed. It measures the number of accessible projector velocities, not the degeneracy D itself. The commutant

$$\mathfrak{C}_X = \{M \in \text{End}(P\mathcal{H}) : [M, A] = 0 \text{ for all } A \in \mathfrak{A}_X\} \quad (23)$$

records exact block structure invisible to an eigenvalue histogram of one curvature matrix. A large commutant signals conserved subblocks or repeated representations. A scalar algebra has the entire endomorphism algebra as its commutant; an irreducible full matrix algebra has only the center. These are algebraic statements and should be established before a random-matrix law is fitted.

The Hermitian curvature combination $i(X_a^\dagger X_b - X_b^\dagger X_a)$ belongs to $\mathfrak{A}_X$. Its centrality is therefore another useful coordinate. We distinguish a fixed projector ($\mathfrak{X} = 0$), central curvature ($F_{ab} \propto P$), structured or reducible response (nonzero $\mathfrak{X}$ with an exact branch or block decomposition), and finite-size stochastic response (noncentral matrices with measured ensemble fluctuations). The last label is empirical and rank-specific. It does not mean that $\mathfrak{A}_X$ has been proved to become a free or invariant random algebra as $D \rightarrow \infty$.

B. Protection class

Constraint-kernel protection occurs when $H = C^\dagger C$ and the common nullity of the constraints is constant. The Laughlin and Moore–Read parents fall in this class, with two- and three-body clustering conditions, respectively. Exactness-preserving transport rotates C together with its kernel; a generic fixed-parent perturbation can instead produce a nonscalar $P(\partial H)P$ and split the fiber. The same protection class consequently permits both a finite exact path and a splitting probe, depending on the tangent.

Cohomological protection occurs for a nilpotent Q and $H = \{Q, Q^\dagger\}$. Harmonic representatives identify the zero modes with Q -cohomology. The random cubic supercharge and the independence-complex lattice model share this origin, but their tangent spaces and sampling units are not the same. The Hodge decomposition adds an exact/coexact branch structure to $\mathfrak{X}$. A Gaussian comparison that ignores that branch structure tests a different null from one that conditions on it.

Stabilizer protection follows from a commuting Pauli constraint algebra and its dependent relations. In X-cube, positive coefficient changes leave the common eigenspace

fixed, whereas unitary conjugation moves it isospectrally. The protected rank is exponentially large in a linear system size, yet the opened transport curvature remains central. This example directly separates the count of protected states from the complexity of their accessible geometry.

A quasi-degenerate isolated cluster, such as the checkerboard replication, belongs to none of these exact classes at finite size. It can test whether a geometric trend survives an internal bandwidth, but it cannot support the algebraic spectral-silence statement used for an exact kernel.

C. Mixing class

The frozen class has $\partial_a P = 0$. X-cube coefficient reweighting is the exact example; changing PHP inside a fixed subspace can also alter spectral statistics without changing the projector. Such a tangent provides no geometric chaos datum, regardless of D .

The central class has $X_a \neq 0$ for some tangents but $F_{ab} = f_{ab}P$ on the protected fiber. X-cube unitary transport realizes this case. It produces a Berry phase density and can have a nonzero quantum metric, but all states in the fiber see the same infinitesimal holonomy. The traceless curvature and its connected variance vanish.

The structured class has a noncentral response subject to a proper invariant decomposition. Local lattice SUSY supplies exact and coexact Hodge branches; structured Laughlin and Moore–Read generator lines restrict the accessible operator span. Here a response rank alone is inadequate. The commutant, branch weights, and channel covariance must be retained in the null model. Failure of a naive isotropic comparison can reflect omitted two-point structure rather than a connected higher moment.

The stochastic class is assigned only when an ensemble of noncentral responses has been measured. The finite Laughlin panels, the Moore–Read fixed-base panels, and random-supercharge realizations exhibit such fluctuations under their registered protocols. Their higher-order status is not uniform: the rank-120 Moore–Read case passes the centered full-cumulant direction; the smaller Moore–Read and lattice cases do not; SUSY SYK rejects a separable Hodge-Gaussian null but lacks a complete entrywise-covariance test. “Finite-size stochastic response” is therefore a mixing description, not a universal closure claim.

D. Statistical class and claim level

Deterministic algebraic controls form the first statistical class. Their results—zero projector derivative, scalar curvature, or a vanishing connected variance—follow from exact identities. They need numerical audits of implementation and sign, but no bootstrap population is created by repeating eigenvalues inside the code space.

Fixed-base tangent populations form the second class. Moore–Read and local lattice SUSY use complete panels treated as exchangeable conditional on a common Hamiltonian. Exact finite-population centering and a training-only whitener prevent the same inference panels from selecting their own null. The resulting jackknife quantifies panel-to-panel variation at that base. It does not quantify variation across disorder, parent Hamiltonians, or topological phases.

Independent disorder realizations form the third class. The random supercharge calculation samples the couplings anew, so one realization is a Hamiltonian-level unit. Its historical sealed prediction is stronger than a same-Hamiltonian resampling statement, but its covariance null is narrower than the full complex Gaussian law. Statistical independence of the units does not repair a misspecified covariance model.

Loop or moduli-space sampling forms a fourth class for global observables. Wilson loops and Chern numbers compare transported frames over extended parameter cycles. Repeated mesh points on one loop are not independent realizations, and the loop parameter is not physical time. These global objects can retain structure even when a local curvature statistic resembles a finite random-matrix reference.

This classification fixes the order of a geometric-chaos claim. One first states the protection mechanism and verifies a constant-rank gap. One then specifies the tangent class, computes the response rank, and identifies the commutant or branch structure of $\mathfrak{A}_X$. Only after that does one choose a statistical class, match the complete relevant two-point structure, and test the higher-order status. An asymptotic Geometric ETH would require these ingredients along a growing-rank sequence with a fixed operator class and controlled connected cumulants. The independent model/operator ensemble has not been established here, and the present finite-rank comparisons do not supply that theorem.

X. DISCUSSION AND LIMITATIONS

A. What the finite-rank comparison establishes

The calculations answer a question that cannot be posed through ordinary level statistics. When a parent Hamiltonian has a rank- D eigenspace pinned to one energy, neither its zero spacings nor its spectral form factor distinguishes a rigid projector from one that rotates strongly under local changes of couplings. The non-Abelian quantum geometric tensor resolves this difference because it measures virtual excursions from the protected fiber into the gapped complement. The relevant randomness is therefore randomness of the embedding $P(\lambda) \subset \mathcal{H}$, not randomness of the repeated eigenvalue E_0 .

Across clustered Hall parents and cohomological kernels, the response is nontrivial even though the protected energies remain equal at the accepted base point or along

an exact transported family. Several finite-rank curvature spectra display Jacobi-like local eigenvalue repulsion. This is the part of the result that most closely resembles an ordinary random-matrix criterion: nearby curvature eigenvalues avoid one another after the finite- D density is treated correctly. It is also the narrowest part of the claim. The comparison does not show that the full response matrix is drawn from an invariant Jacobi ensemble. Its channel covariance, pseudocovariance, and exact/coexact or stabilizer branch structure remain visible at larger scales.

We use *geometric chaos* for this combination of non-commuting projector response, stochastic local curvature correlations, and a nontrivial complement channel. The phrase does not assert that the parent Hamiltonian thermalizes. Indeed, the protected sector has no internal energy differences on which a conventional ETH ansatz could act. What is measured is the sensitivity and correlation structure of the degenerate subspace under a declared tangent ensemble. This meaning is consistent with the use of eigenstate deformation as a chaos-sensitive observable [8] and with the proposal to use Berry curvature inside BPS multiplets [9], while remaining agnostic about long-time relaxation.

B. Protection, mixing, and the meaning of a tangent

The separation between S_a and X_a is more than a book-keeping device. S_a asks whether a perturbation resolves energies inside the fiber; $X_a = Q(\partial_a P)P$ asks whether it changes the fiber's embedding. An exactly transported parent can have $S_a = 0$ and large X_a , whereas a scalar rescaling can have both responses trivial. Conversely, an arbitrary local potential can split the fiber and rotate its spectral cluster at the same time. Calling all three operations *deformations of the parent* would erase the physical distinction on which the geometric test rests.

Accordingly, finite-path claims in this work come only from exactness-preserving transport. For the Laughlin and Moore–Read generator panels, the constraint map is transported unitarily, the kernel rank remains fixed, and the differentiated identity gives $S_a = 0$. Closed paths can then support a Wilczek–Zee holonomy [6]; its non-Abelian part contains information not fixed by the first Chern number. In a gapped topological phase, quasiadiabatic continuation supplies the broader setting for such projector motion [11], although our exact parent paths are more restrictive than a generic phase-preserving deformation.

The Laughlin local-potential panels are fixed-parent base-point probes. Their $Q(\partial_a H)P$ matrices are well defined because the rank- D spectral cluster is isolated at the parent, but a generic finite potential does not preserve exact equality of its eigenvalues. These panels can test local response statistics at the exact point; they cannot support a claim about a globally degenerate bundle away from it. This boundary is especially important when interpreting a large curvature response: sensitivity of an

isolated cluster is not by itself protection of a degenerate multiplet.

C. Local Jacobi behavior and global memory

A finite Jacobi process is an appropriate reference for eigenvalues generated by the relative orientation of random subspaces [10]. Agreement with its local repulsion law tests correlations within a single whitened curvature matrix. It does not test every index of X_a , nor does it remove the covariance between tangent directions. The analogy with standard spectral chaos must therefore stop before one equates a local point process with a complete matrix distribution.

This separation explains why Jacobi-like local eigenvalue repulsion can coexist with global covariance memory. A clustered parent restricts which complement states are reached by a local generator. A supercharge resolves the response into exact and coexact branches. A stabilizer algebra confines the response to a small commutant-determined sector. These restrictions survive covariance whitening unless the whitening includes the complete entry covariance and pseudocovariance with the correct independent sampling unit. They can also generate fourth-order structure that is invisible in a nearest-neighbor spacing histogram.

For that reason, the four-point analysis uses all three complex Wick pairings and treats the full fixed-Hamiltonian tangent panel as one statistical object. Matrix entries and pairs of labels within that panel are not independent realizations. A nonzero raw connected estimate obtained by resampling them as if they were independent would measure pseudoreplication as readily as non-Gaussian physics. The final evidence registry therefore uses the exactly enumerated population mean of each registered finite ensemble, a channel whitener fitted on a frozen 12-panel training split, and the complete three-pairing statistic on 12 held-out panels. The primary inference treats those held-out panels as exchangeable conditional on the fixed base Hamiltonian, uses a Student- t_{11} reference, and applies a Bonferroni familywise correction over the five registered Moore–Read and lattice-SUSY cases. The reverse 12/12 split is descriptive and is not pooled with the primary inference.

Only the registered finite-rank Moore–Read $N = 6$ case passes this corrected gate: its directional estimate is 0.18490062, and its familywise one-sided lower bound is 0.039160542 > 0. The Moore–Read $N = 4$ lower bound remains negative, as do those for lattice-SUSY ranks $D = 2, 4, 8$. No positive complete-cumulant statement is made for those nonpassing cases, and nonpassing is not evidence for Gaussianity. The Laughlin and random-SYK production data have not been evaluated with this corrected statistic. The $N = 6$ result is therefore evidence for a population-centered fourth-order residual in one finite-rank, fixed-base tangent ensemble; it is not evidence for an independent-Hamiltonian ensemble or a common

large-rank law.

D. What is mechanism dependent

The cross-mechanism comparison contains both positive and negative cases. Two- and three-body clustering produce large common kernels whose response is set by forbidden coincidence channels. Random-supercharge cohomology gives a dense coupling-space ensemble with exact and coexact Hodge branches. The local lattice supercharge preserves cohomology but has a different locality and a much smaller sequence of protected ranks. These systems need not share a covariance tensor, and the present data do not support pooling their uncertainties into one scaling curve.

The X-cube model is the decisive structured control. Changing coefficients of fixed commuting stabilizers leaves the eigenprojector unchanged, so the coefficient-space curvature is zero. Under a specified local unitary transport the projector moves and its curvature can be nonzero, but the curvature eigenvalue is deterministic and its connected variance vanishes. This example separates three propositions that are sometimes conflated: macroscopic exact degeneracy, nonzero geometric response, and stochastic geometric chaos. Neither of the first two implies the third.

The continuum and Kapit–Mueller Laughlin calculations give a second kind of control. They share two-body clustering but differ microscopically. Agreement of a local statistic across them is evidence that the effect is not solely a lattice artifact; disagreement of a global covariance contraction is allowed because the tangent realization and complement spectrum differ. Parent dependence is consequently part of the result rather than an error bar to be averaged away.

E. Relation to Geometric ETH

For the finite systems studied here, *Geometric ETH* is an organizing name for a hierarchy of tests, not a theorem. The weakest level is informative geometry under exact degeneracy. A stronger, domain-limited level adds local random-matrix-like correlations for a specified protection mechanism and tangent class. An asymptotic statement would require a sequence with growing protected rank, a fixed normalization and operator class, controlled external gaps, converged local statistics, and decay or convergence of the covariance-whitened connected tensors. None of these requirements can be replaced by increasing the ambient Hilbert-space dimension alone.

Our result is finite-rank, cross-mechanism evidence. It is not an asymptotic Geometric ETH theorem and not a universal Geometric ETH theorem. In particular, an independent model/operator class has not been established: the available sequences change either the protection algebra, the tangent panel, or both. We also do not demonstrate thermalization, a Lyapunov exponent, an

out-of-time-order growth regime, or equivalence to the usual ETH ansatz for few-body observables. Those are distinct dynamical propositions.

These limitations indicate what would raise the claim. Within one mechanism, one must extend a fixed operator prescription through a substantially larger rank sequence and preserve a positive external gap. Independent Hamiltonian or disorder realizations must replace fixed-base panels wherever an ensemble claim requires them. Complete covariance and pseudocovariance must be estimated with uncertainty at that realization level. Only after these conditions are met would it be meaningful to test a common large- D law. Until then, mechanism-complete sampling is more informative than adding an unbounded list of models with incompatible tangent definitions.

F. Conclusion

Exactly degenerate energies do not make quantum chaos unobservable; they move the question from eigenvalue spacings to the geometry of the protected projector. The response $X_a = Q(\partial_a P)P$ remains finite when the intrafiber splitting channel vanishes, and its non-Abelian curvature separates rigid protected fibers from stochastically mixing ones. Across clustered Hall parents, random and local cohomological models, and the X-cube control, we find finite-rank, cross-mechanism evidence for a hierarchy in which locally Jacobi-like curvature can retain parent- and operator-dependent covariance at larger scales.

The hierarchy is the main conclusion. Local repulsion is not full Gaussian closure; nonzero curvature is not stochasticity; base-point sensitivity is not finite-path protection; and exact degeneracy is not thermalization. The negative stabilizer result is as important as the stochastic cases because it shows that exact protection does not automatically generate a random geometry. The positive statements are therefore conditional on the declared mechanism, tangent class, protected rank, and statistical unit.

The present evidence supports a finite-size, domain-limited use of the term Geometric ETH. It leaves the asymptotic problem open. Establishing that stronger statement will require growing-rank sequences with an unchanged operator prescription and independent-ensemble covariance closure. Results from the corrected full-cumulant gate are restricted to Moore–Read $N = 6$; Moore–Read $N = 4$ and lattice-SUSY $D = 2, 4, 8$ do not pass, while Laughlin and random SYK remain untested by that statistic. This finite-rank result does not establish an asymptotic or universal Geometric ETH. Within these boundaries, non-Abelian quantum geometry provides information precisely where the degenerate spectrum does not.

ACKNOWLEDGMENTS

Numerical data, exact-rank audits, frozen inference splits, and source hashes accompany the reproducibility package. We thank the Quantum Harness community for discussions of numerical validation and the scope of the claims.

-
- [1] M. Srednicki, Chaos and quantum thermalization, *Physical Review E* **50**, 888 (1994).
 - [2] L. D'Alessio, Y. Kafri, A. Polkovnikov, and M. Rigol, From quantum chaos and eigenstate thermalization to statistical mechanics and thermodynamics, *Advances in Physics* **65**, 239 (2016).
 - [3] X. Chen and A. W. W. Ludwig, Universal spectral correlations in the chaotic wave function, and the development of quantum chaos, *Physical Review B* **98**, 064309 (2018), [arXiv:1710.02686 \[cond-mat.str-el\]](#).
 - [4] T. Kato, On the adiabatic theorem of quantum mechanics, *Journal of the Physical Society of Japan* **5**, 435 (1950).
 - [5] M. V. Berry, Quantal phase factors accompanying adiabatic changes, *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* **392**, 45 (1984).
 - [6] F. Wilczek and A. Zee, Appearance of gauge structure in simple dynamical systems, *Physical Review Letters* **52**, 2111 (1984).
 - [7] J. P. Provost and G. Vallée, Riemannian structure on manifolds of quantum states, *Communications in Mathematical Physics* **76**, 289 (1980).
 - [8] M. Pandey, P. W. Claeys, D. K. Campbell, A. Polkovnikov, and D. Sels, Adiabatic eigenstate deformations as a sensitive probe for quantum chaos, *Physical Review X* **10**, 041017 (2020), [arXiv:2004.05043 \[cond-mat.stat-mech\]](#).
 - [9] Y. Chen, S. Colin-Ellerin, O. Mamroud, and K. Papadodimas, Chaos of berry curvature for bps microstates (2026), [arXiv:2604.23287 \[hep-th\]](#).
 - [10] B. Collins, Product of random projections, jacobi ensembles and universality problems arising from free probability, *Probability Theory and Related Fields* **133**, 315 (2005).
 - [11] M. B. Hastings and X.-G. Wen, Quasiadiabatic continuation of quantum states: The stability of topological ground-state degeneracy and emergent gauge invariance, *Physical Review B* **72**, 045141 (2005).
 - [12] R. B. Laughlin, Anomalous quantum hall effect: An incompressible quantum fluid with fractionally charged excitations, *Physical Review Letters* **50**, 1395 (1983).
 - [13] B. A. Bernevig and F. D. M. Haldane, Model fractional quantum hall states and jack polynomials, *Physical Review Letters* **100**, 246802 (2008).
 - [14] L. Chen and A. Seidel, Algebraic approach to the study of zero modes of haldane pseudopotentials, *Physical Review B* **91**, 085103 (2015).
 - [15] E. Kapit and E. Mueller, Exact parent hamiltonian for the quantum hall states in a lattice, *Physical Review Letters* **105**, 215303 (2010).
 - [16] G. Moore and N. Read, Nonabelions in the fractional quantum hall effect, *Nuclear Physics B* **360**, 362 (1991).
 - [17] F. Zhang, M. Schossler, A. Seidel, and L. Chen, From frustration-free parent hamiltonians to off-diagonal long-range order: Moore-read and related states in second quantization, *Physical Review B* **108**, 075142 (2023).
 - [18] W. Fu, D. Gaiotto, J. Maldacena, and S. Sachdev, Supersymmetric sachdev-ye-kitaev models, *Physical Review D* **95**, 026009 (2017), [arXiv:1610.08917](#).
 - [19] P. Fendley, K. Schoutens, and J. de Boer, Lattice models with $\mathcal{N} = 2$ supersymmetry, *Physical Review Letters* **90**, 120402 (2003), [arXiv:hep-th/0210161](#).
 - [20] L. Huijse and K. Schoutens, Supersymmetry, lattice fermions, independence complexes and cohomology theory, *Advances in Theoretical and Mathematical Physics* **14**, 643 (2010), [arXiv:0903.0784](#).
 - [21] S. Vijay, J. Haah, and L. Fu, Fracton topological order, generalized lattice gauge theory, and duality, *Physical Review B* **94**, 235157 (2016), [arXiv:1603.04442](#).
 - [22] K. Sun, Z.-C. Gu, H. Katsura, and S. Das Sarma, Nearly flatbands with nontrivial topology, *Physical Review Letters* **106**, 236803 (2011), [arXiv:1012.5864 \[cond-mat.str-el\]](#).
 - [23] N. Regnault and B. A. Bernevig, Fractional chern insulator, *Physical Review X* **1**, 021014 (2011), [arXiv:1105.4867 \[cond-mat.str-el\]](#).